Malware Analysis

By: Oluwatoni Oderinlo
Professor – Dimple Chauhan

Agenda

- Introduction about malware and its types
- Malware analysis process
- Project goal and requirements
- Steps for projects:
  - Anti-malware scanning and find the signature for sample malware using virus total
  - Finding the string values of sample malware using PEViewer tool
  - Malware is packed or unpacked with packed malware form information using PEiD
  - Finding malware implementation and date using PE-Explorer
  - Finding dissembler result using PE-Explorer
  - List of Import functions using Dependency scanner option of PE-Explorer
- Conclusion
- Ref.
- Thank You

Warning

- The information provided in this presentation is only intended for educational purpose only.
- Any collected data during testing must be handled with strict confidentiality and used only for the purposes outlined in this project.
- Unauthorized scanning or exploitation of system is illegal and unethical

Introduction about Malware and its types

The objective of this project is to perform malware analysis on sample malware files using tools like virus total, PEViewer Tools, Pe-Explorer, PEID understanding and defending against these diverse threats.
Malware (short for malicious software) is any program or code designed to harm, exploit, or disrupt systems, networks, or devices. Cybercriminals use malware to steal sensitive data, gain unauthorized access, sabotage systems, or monitor user activity. Malware is a significant threat in the cybersecurity domain, requiring robust defense and analysis techniques to mitigate its impact.

Cont. Introduction about Malware and its types

Types of Malware

Viruses: Self-replicating programs that infect files and spread when executed

Worms: Standalone programs that spread across networks without user intervention.

Trojan Horses: Malicious software disguised as legitimate applications to trick users into installation.

Ransomware: Encrypts data and demands payment to restore access

Spyware: Secretly monitors user activity and collects sensitive information

Keyloggers: Record keystrokes to steal credentials and sensitive data.

Malware Analysis process

Anti-malware Scanning:
•Perform an initial scan of the sample malware using **Virus Total** to identify its signature and check for known threats across multiple anti-virus engines

String Analysis:
•Use the **PEViewer tool** to extract **string values** from the malware sample. This can reveal hardcoded file paths, IP addresses, or malicious commands embedded in the executable.

Packed or Unpacked Analysis:
•Check if the malware is packed using **PEiD**. If packed, note the packer type and consider unpacking the malware for deeper analysis.

Cont. Malware Analysis process

Metadata Inspection:
•Analyze the **implementation date** and other metadata using **PE-Explorer** to determine when and how the malware was created.

Disassembly:
•Use the **Disassembler** feature in **PE-Explorer** to reverse-engineer the malware and inspect its low-level code. This helps in understanding its behavior and potential exploits.

Import Functions Analysis:
•List all **imported functions** using the **Dependency Scanner** in **PE-Explorer**. This step identifies external libraries or APIs that the malware relies on for its malicious activities.

Project goals and requirements

This project will be done in a Virtual environment network and machine
The primary system is Windows server 2022.

Project Goals:

- Understand Malware Behavior
- Identify Malware Signatures:
- Analyze Malware Structure
- Extract Import Functions
- Evaluate Disassembly Results
- Enhance Threat Response

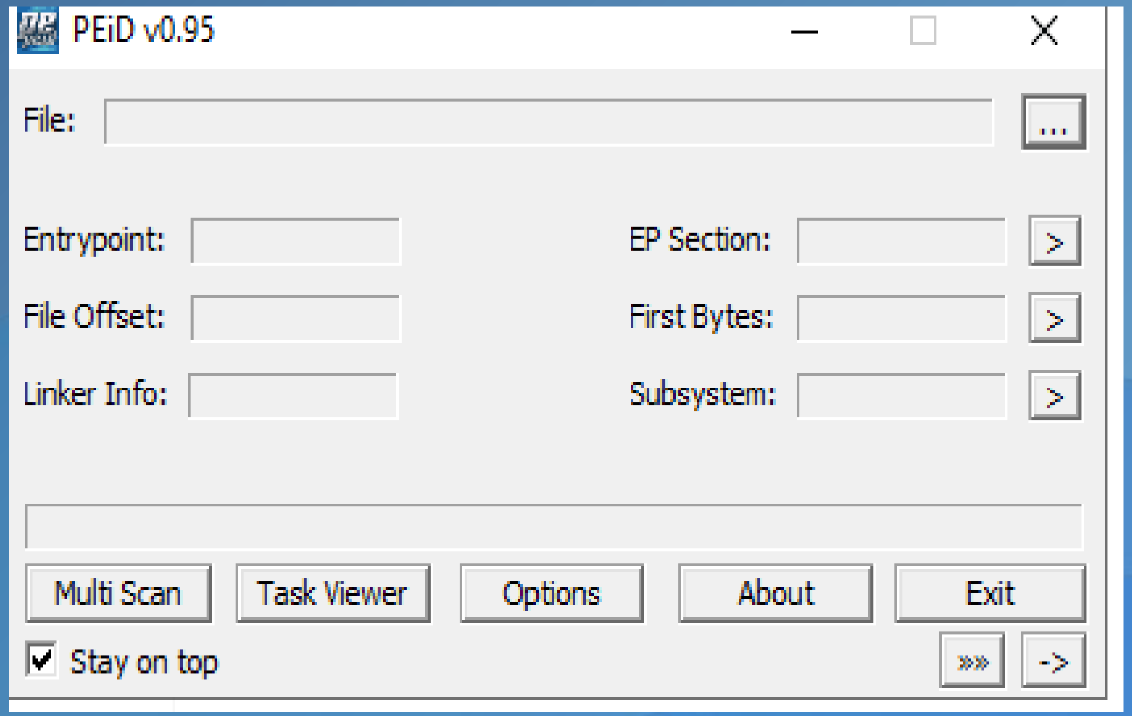
Tools used:

•Virus Total
•PEViewer
•PEiD
•PE-Explorer

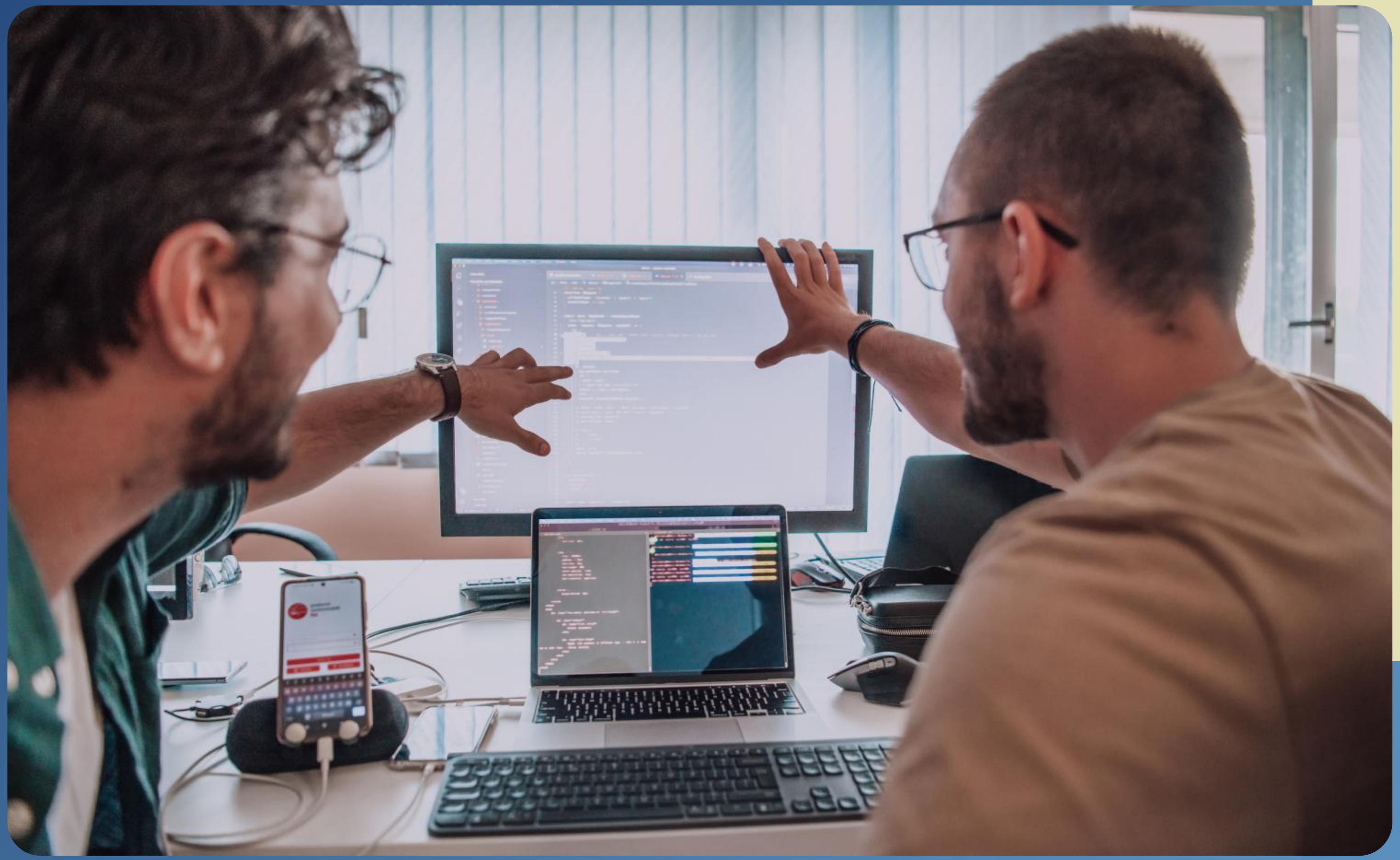
Sample Malware:

Lab01:
Lab02-04

Cont. Project goals and requirements



Steps for Project

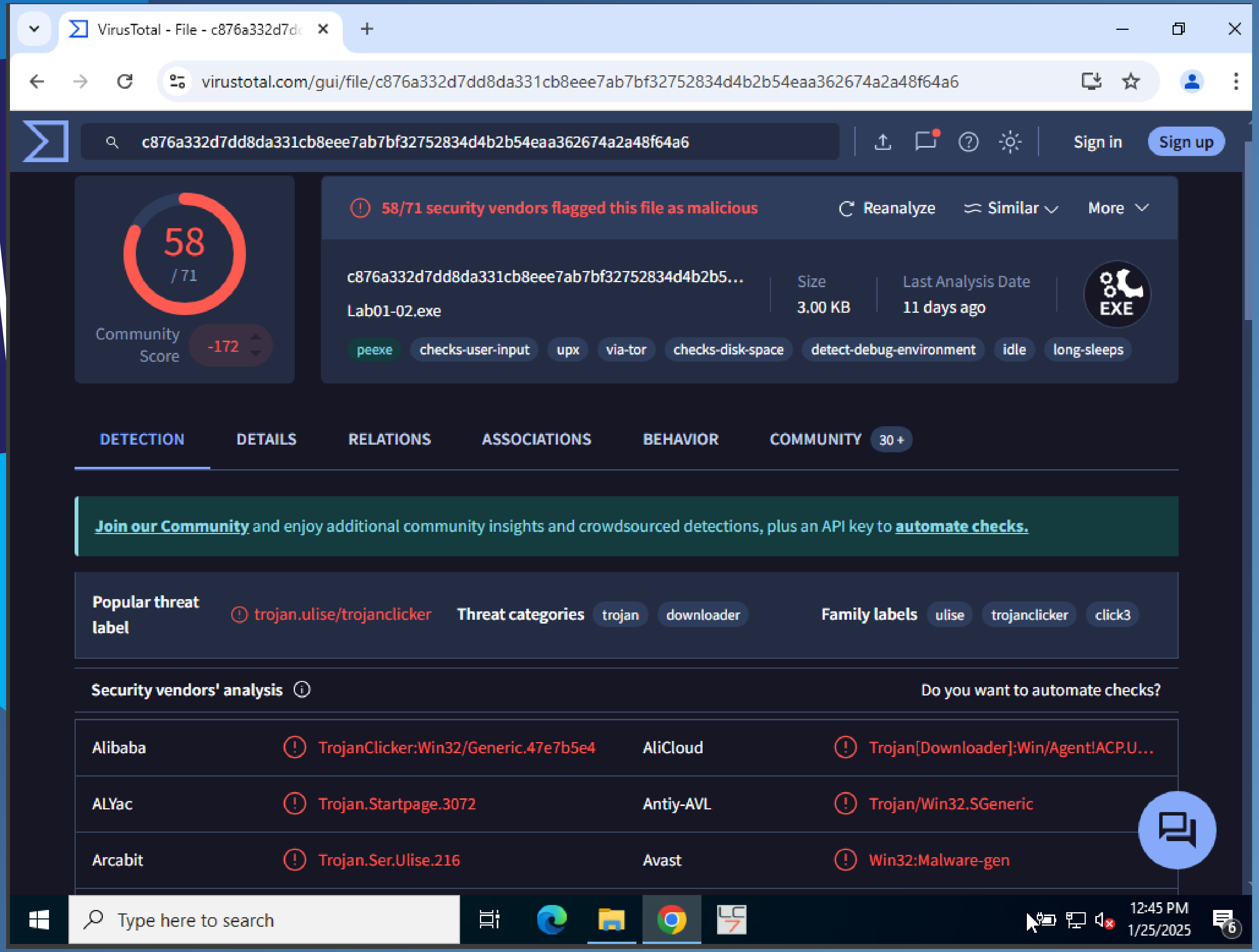


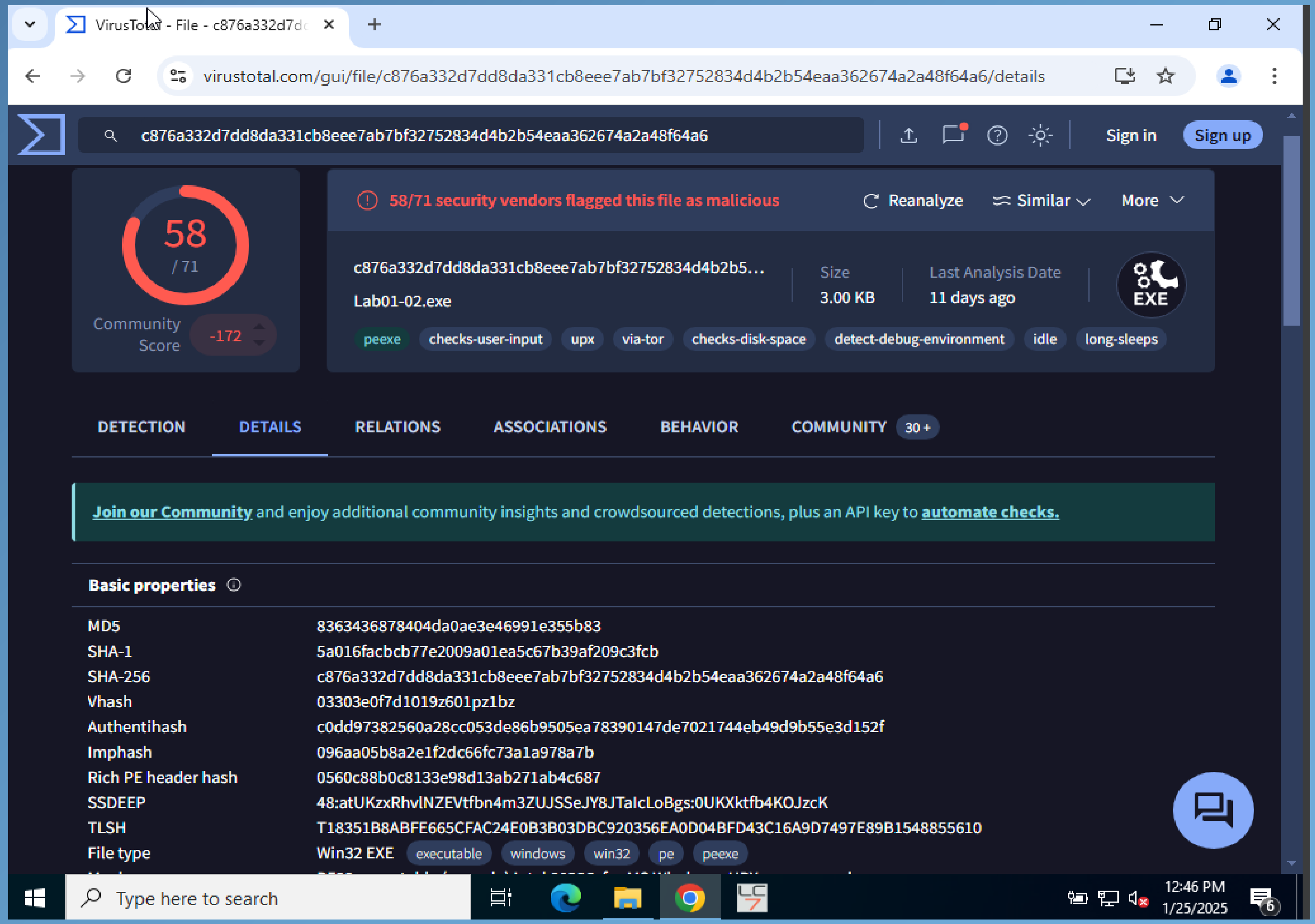
1. Anti-malware scanning and find the signature for sample malware using virus total
2. Finding the string values of sample malware using PEViewer tool
3. Malware is packed or unpacked with packed malware form information using PEiD
4. Finding malware implementation and date using PE-Explorer
5. Finding dissembler result using PE-Explorer
6. List of Import functions using Dependency scanner option of PE-Explorer

Step 1: Lab01-02

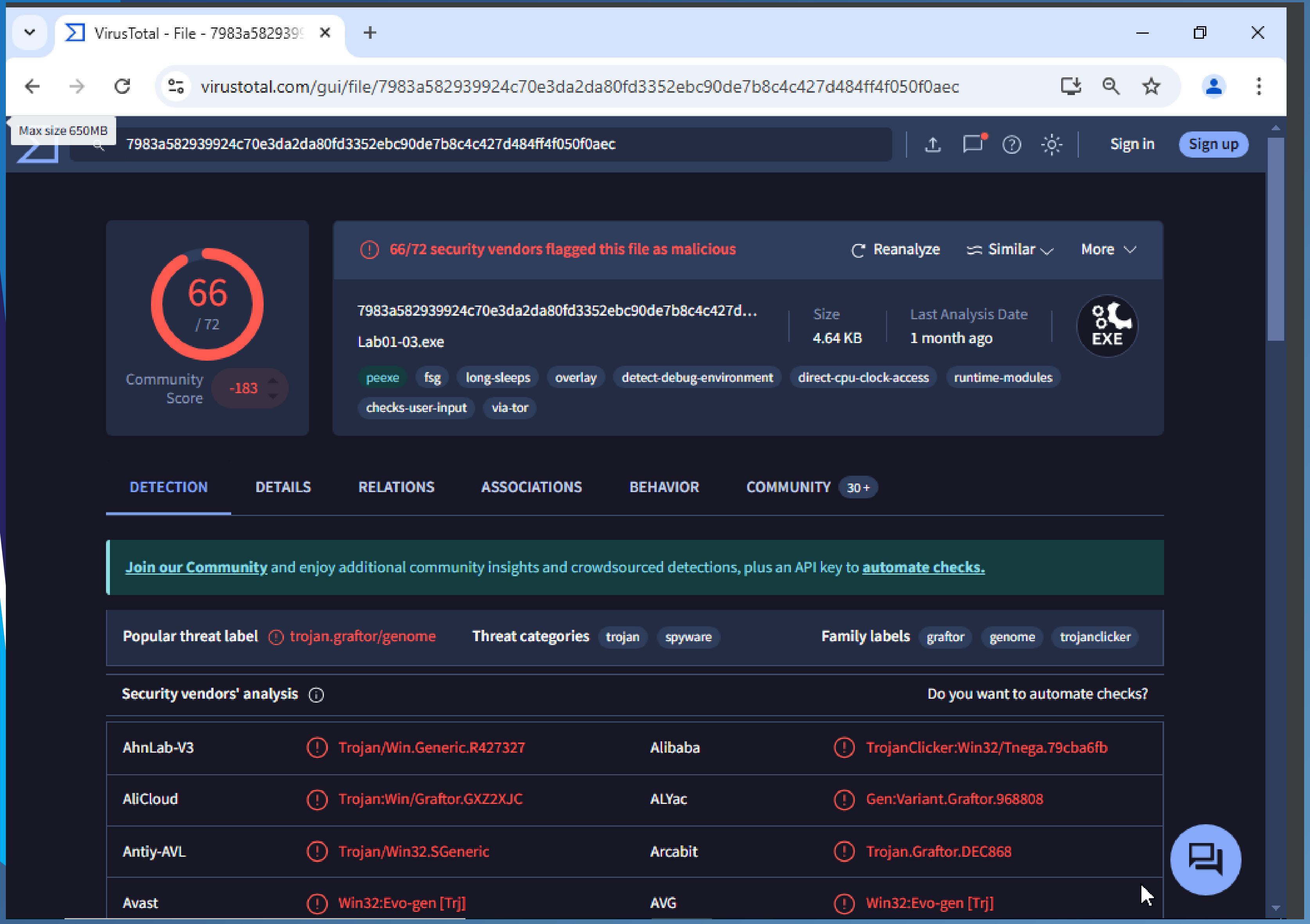
All the screenshots in Step 1 for the sample malware Lab02-04 is the result of the anti-

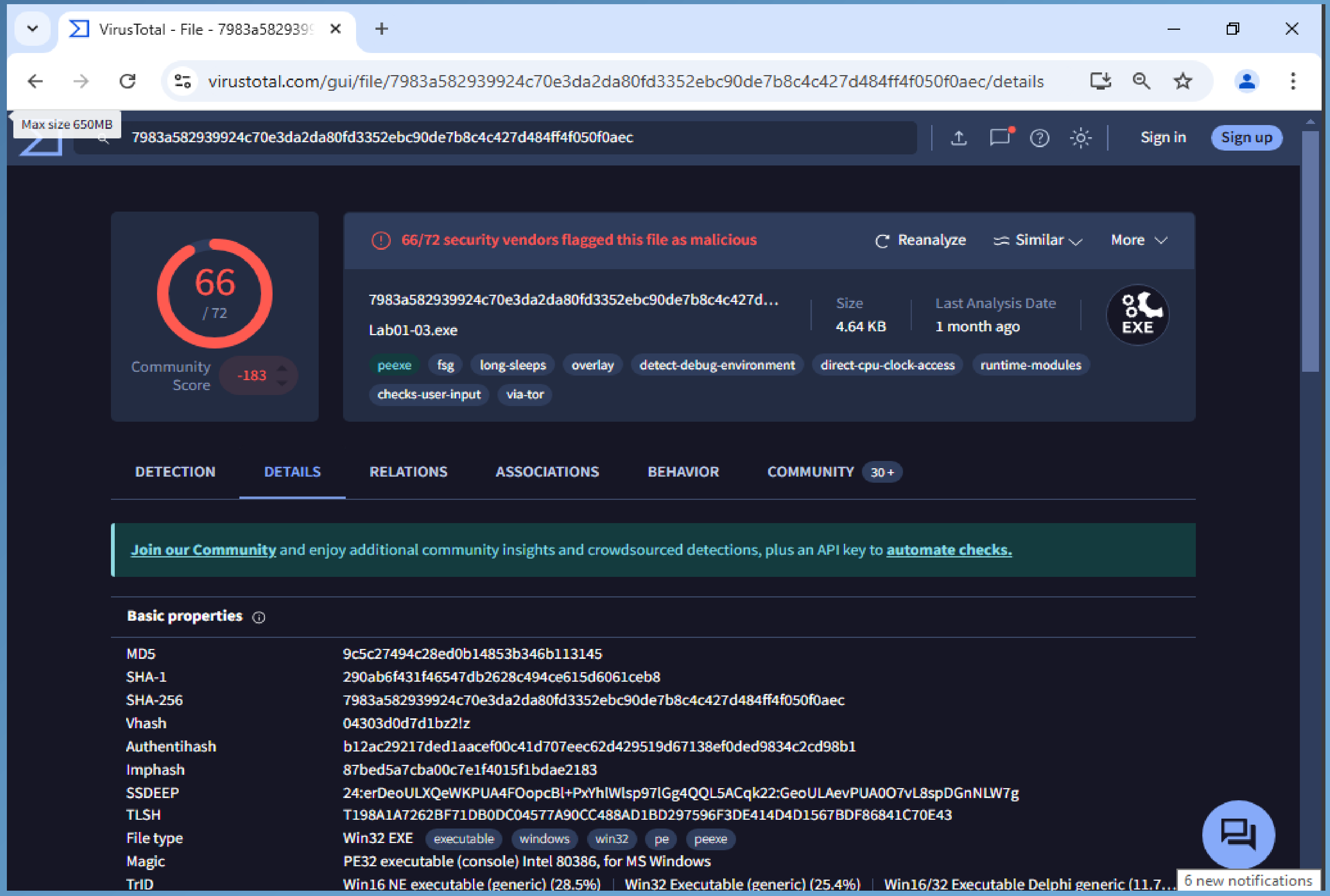
malware scan and the signature associated with the virus using virus total.



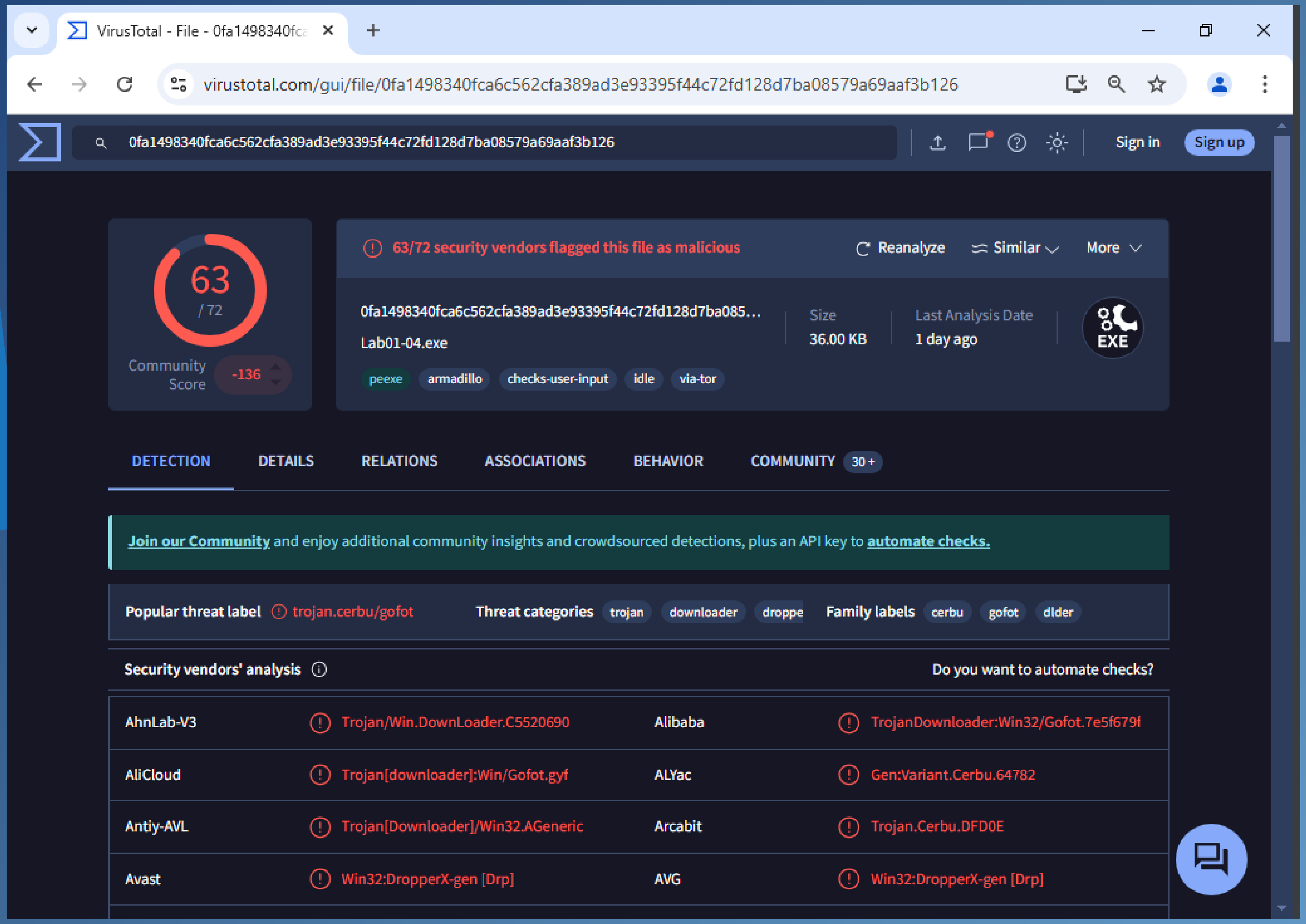


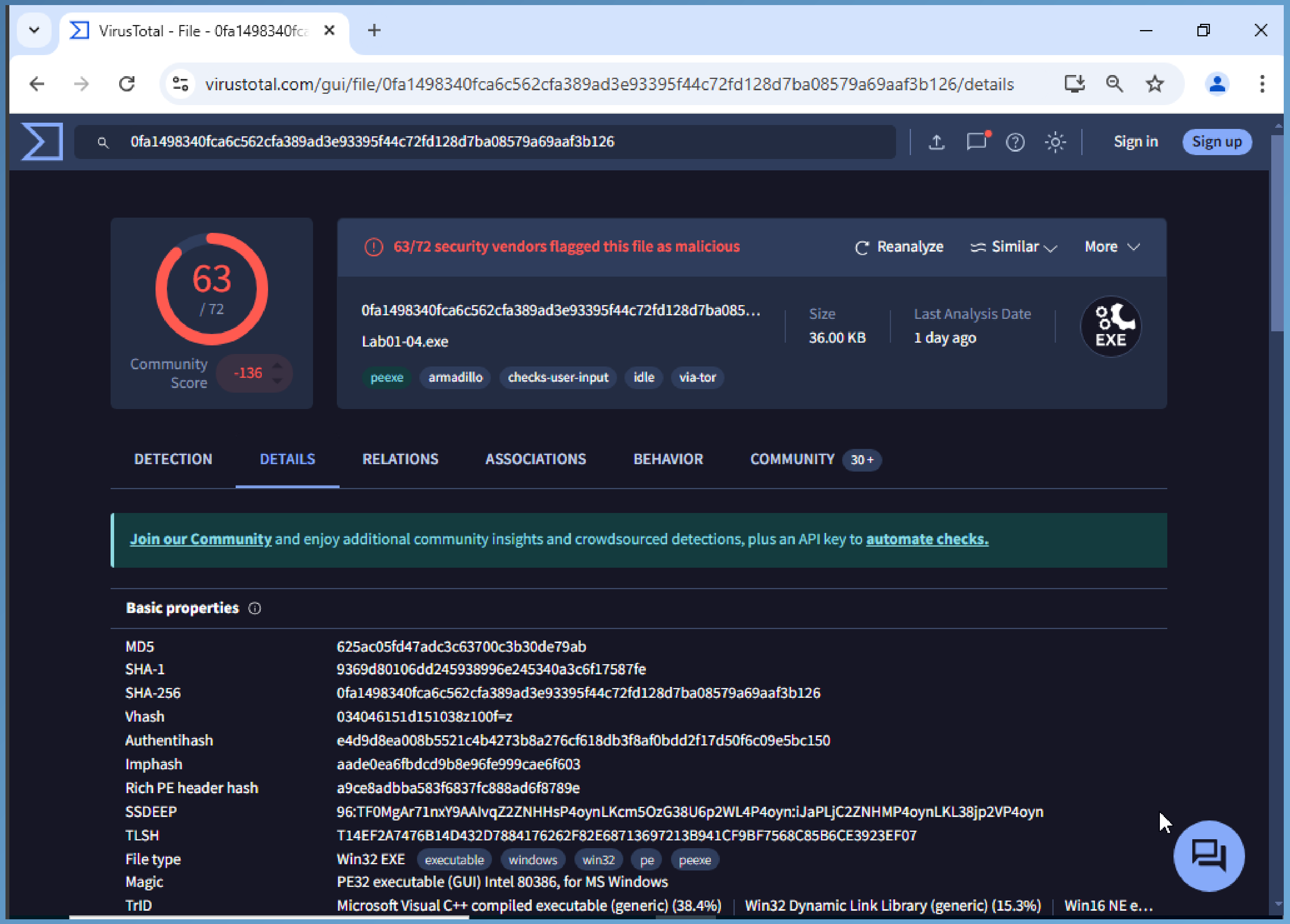
Step 1: Lab01-03





Step 1: Lab01-04

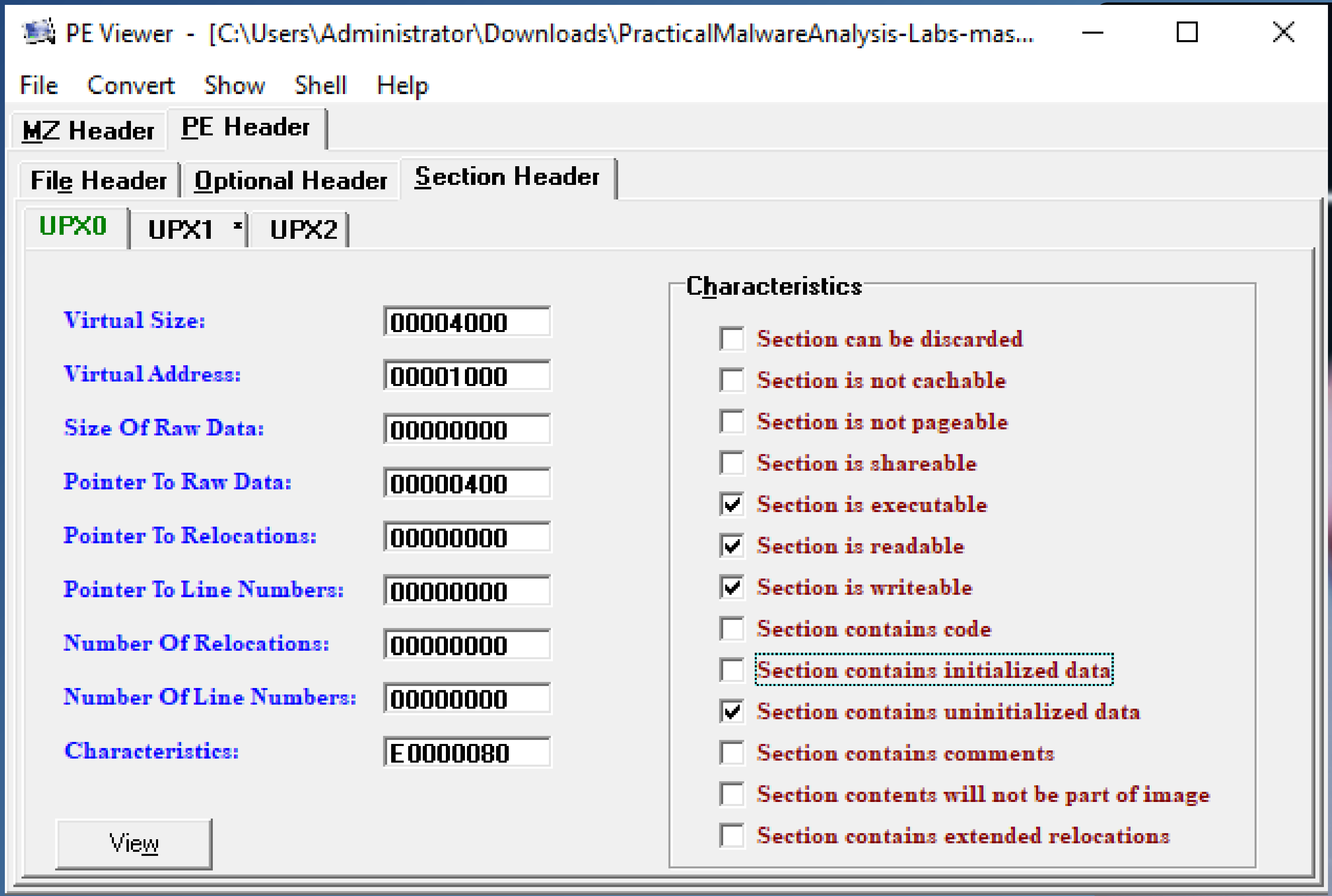


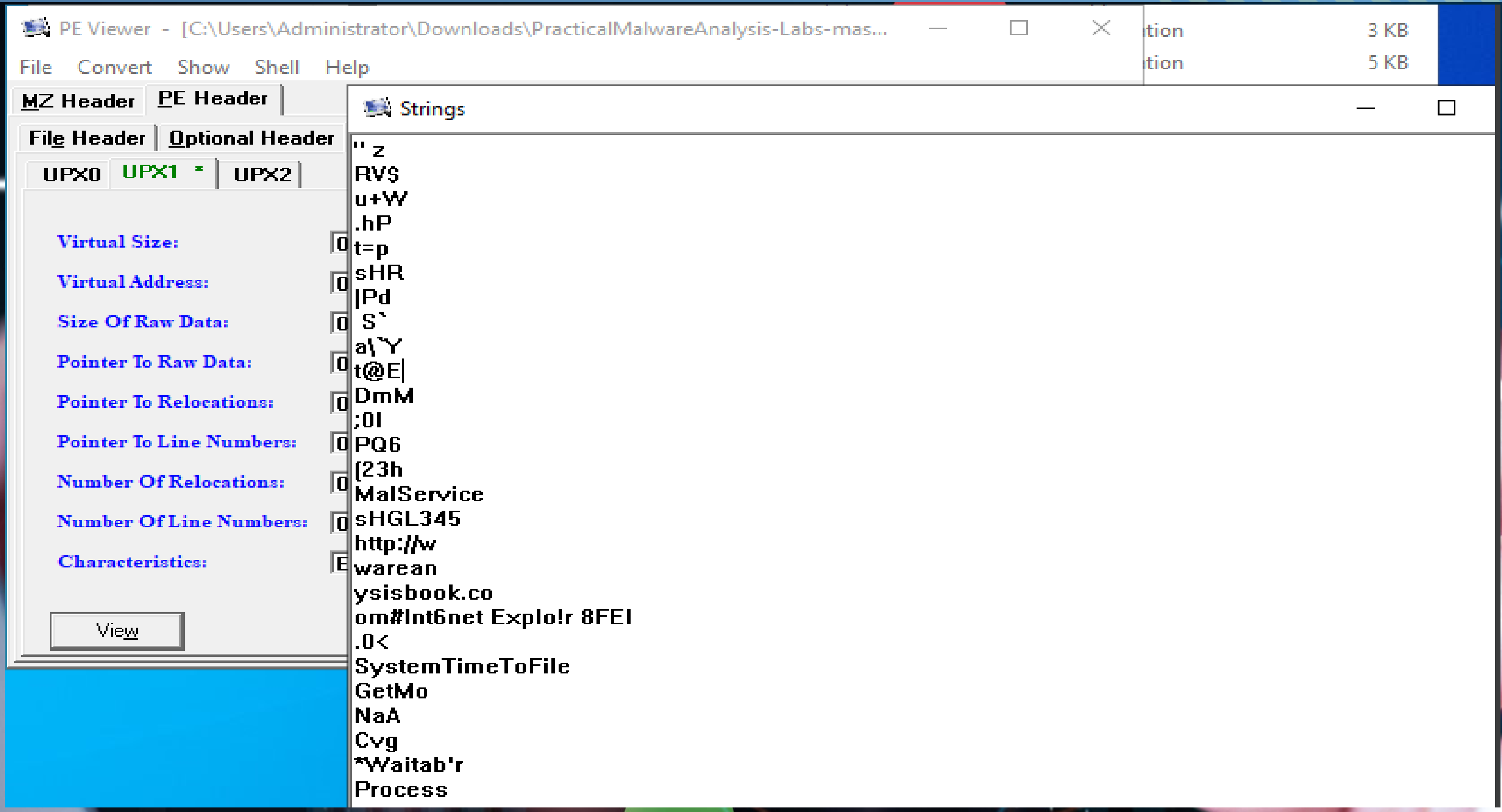


Step 2: Lab01-02

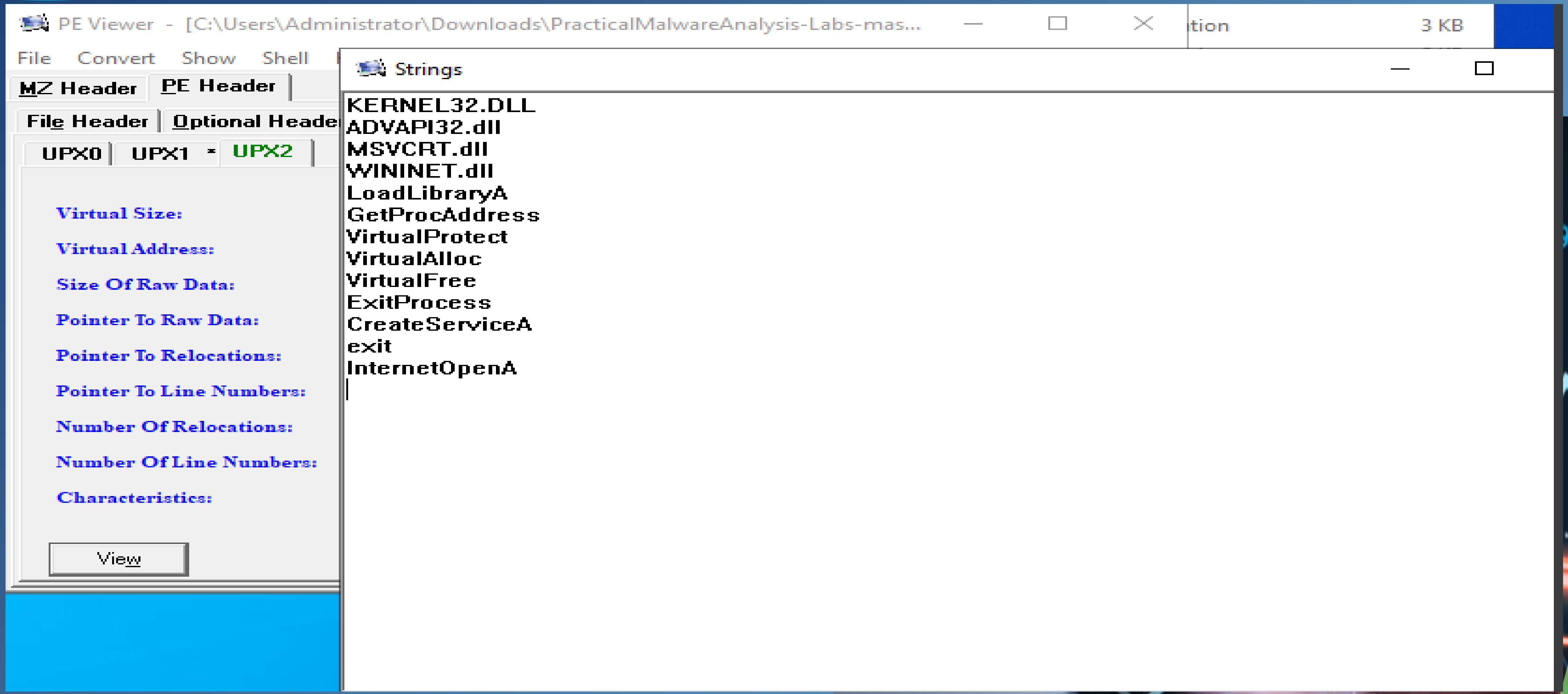
All the screenshots in Step 2 for the sample malware Lab02-04 are the results of using PE Viewer to discover the string values, which are

found under Pe-Header-Section Header

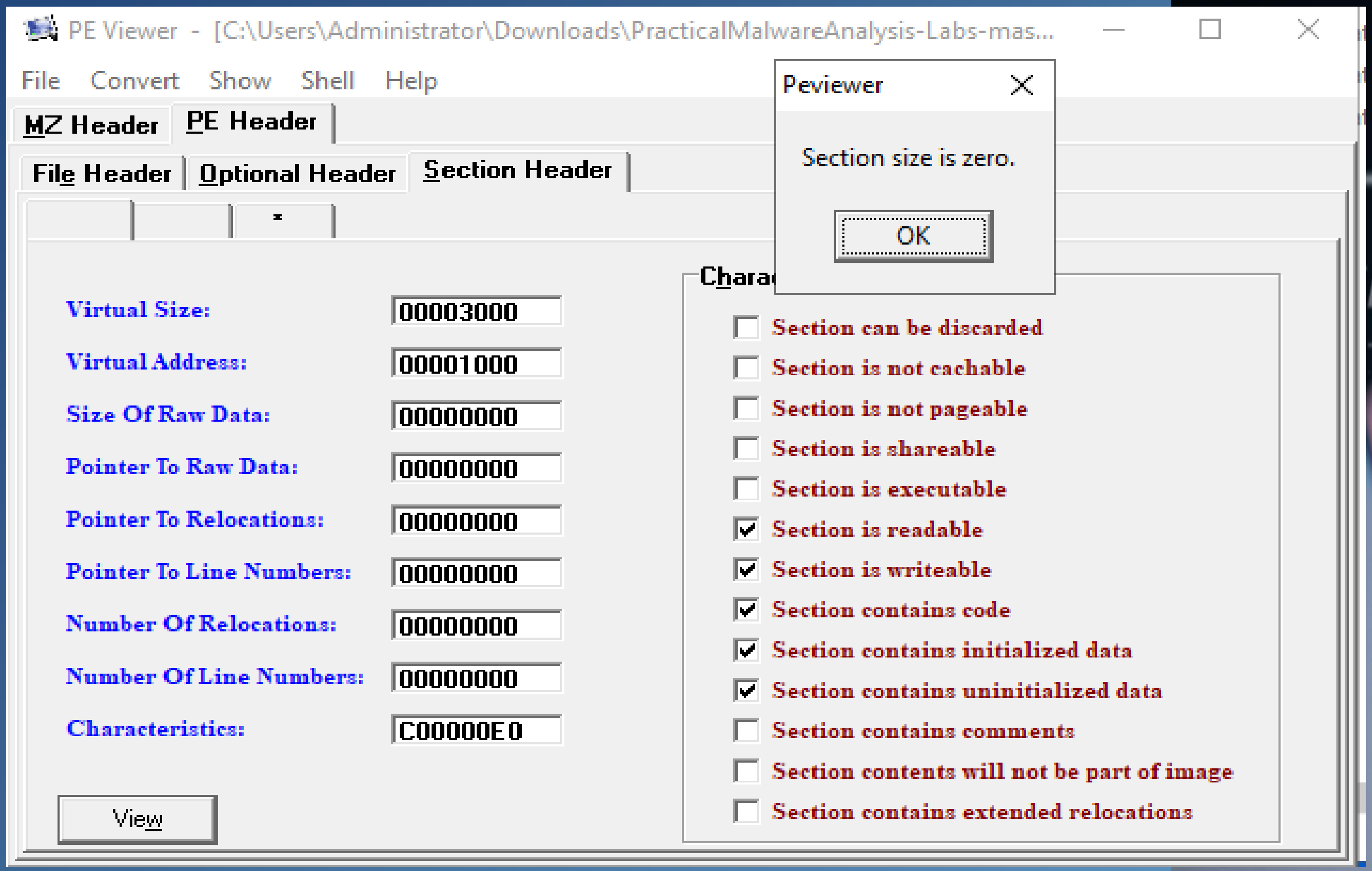


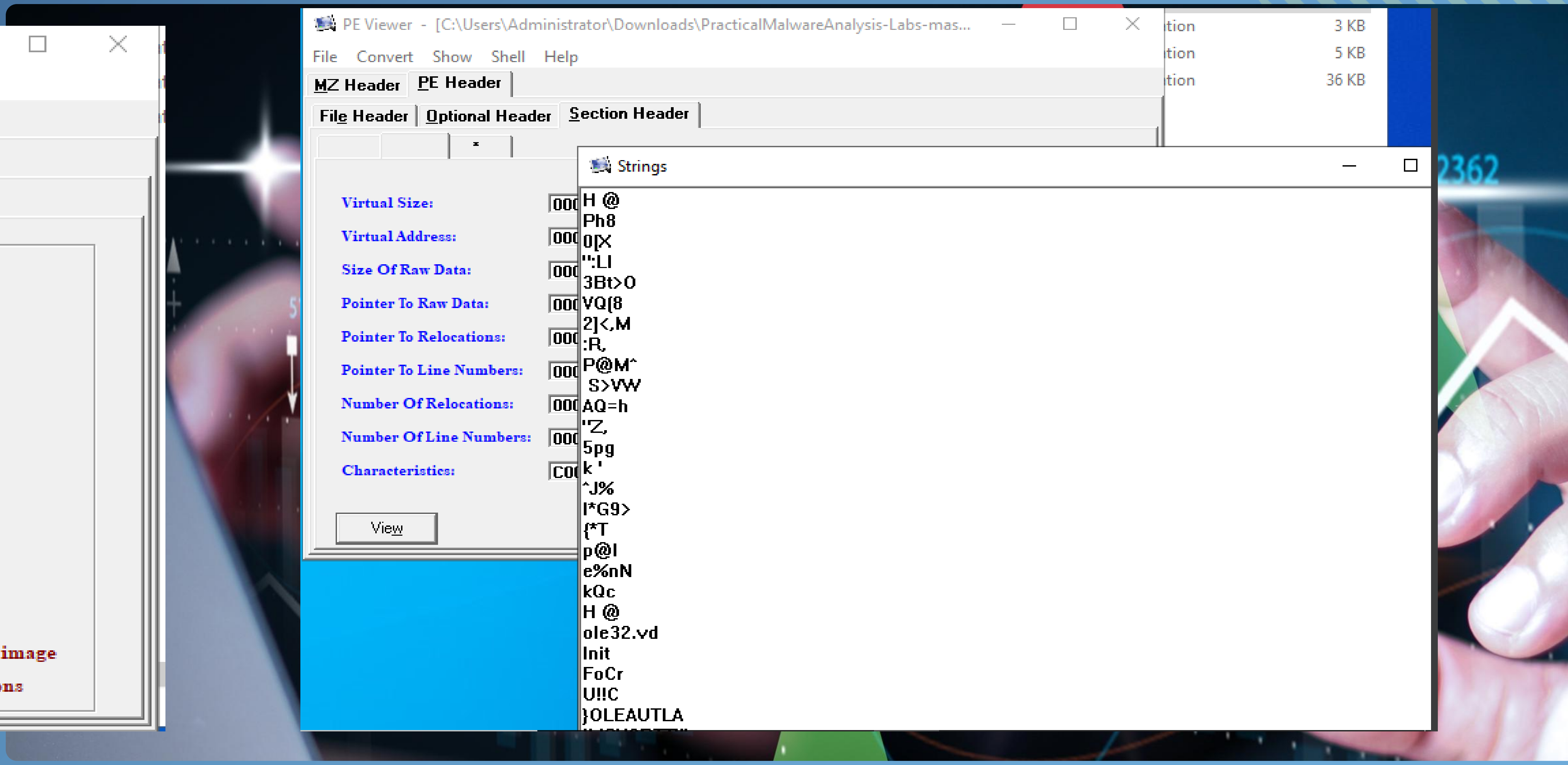


Cont. Step 2: Lab01-02

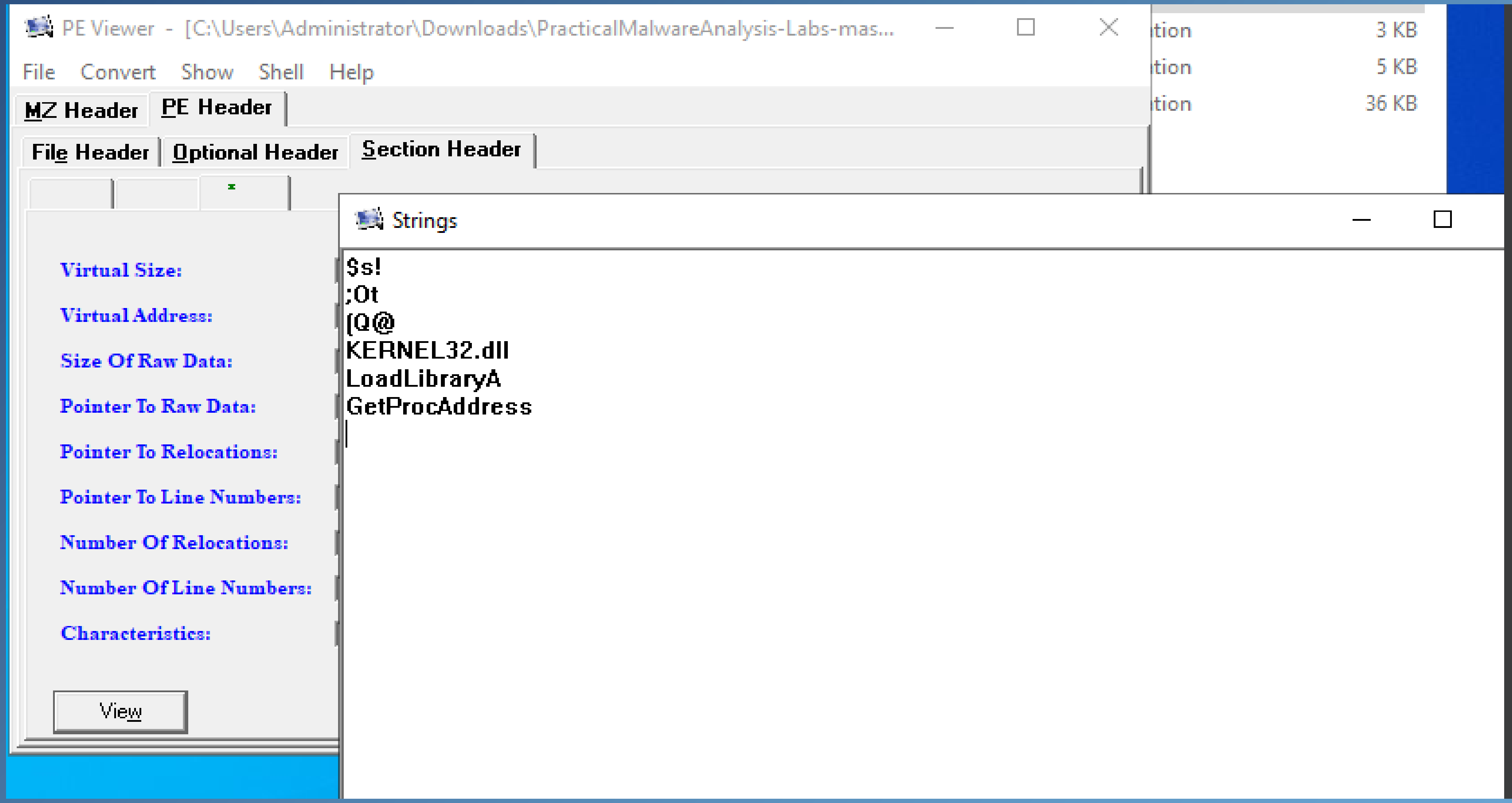


Step 2: Lab01-03

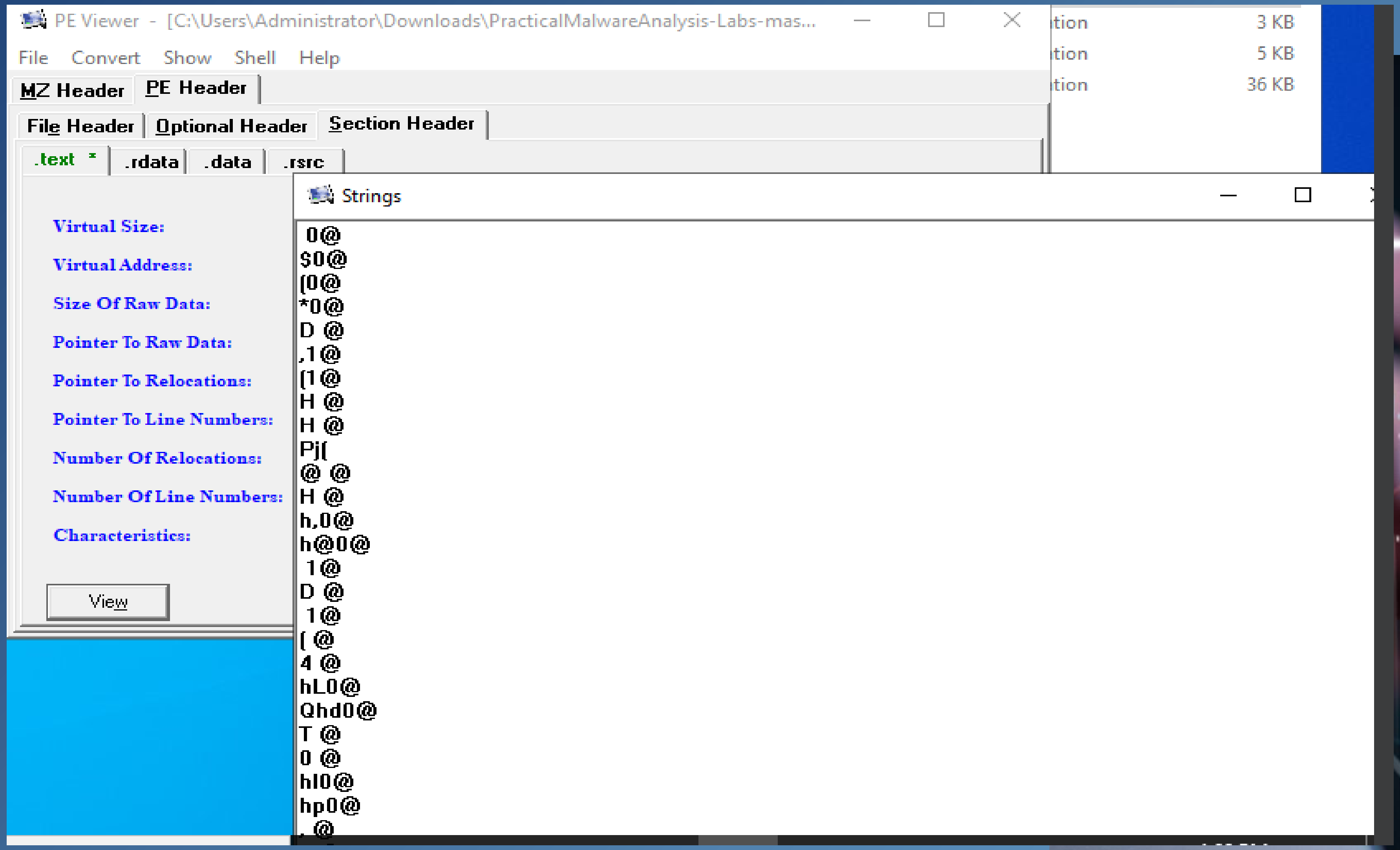


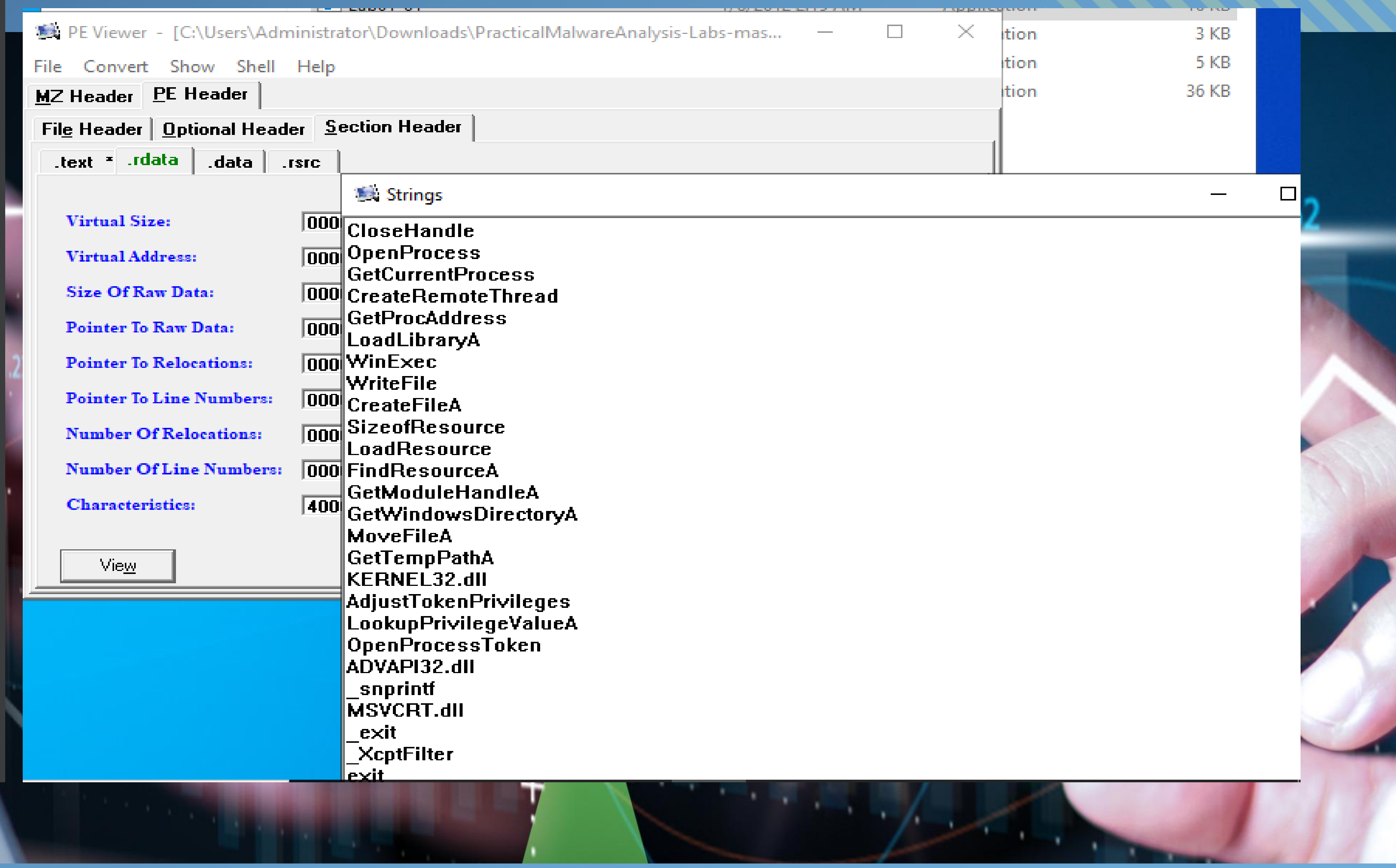


Cont. Step 2: Lab01-03

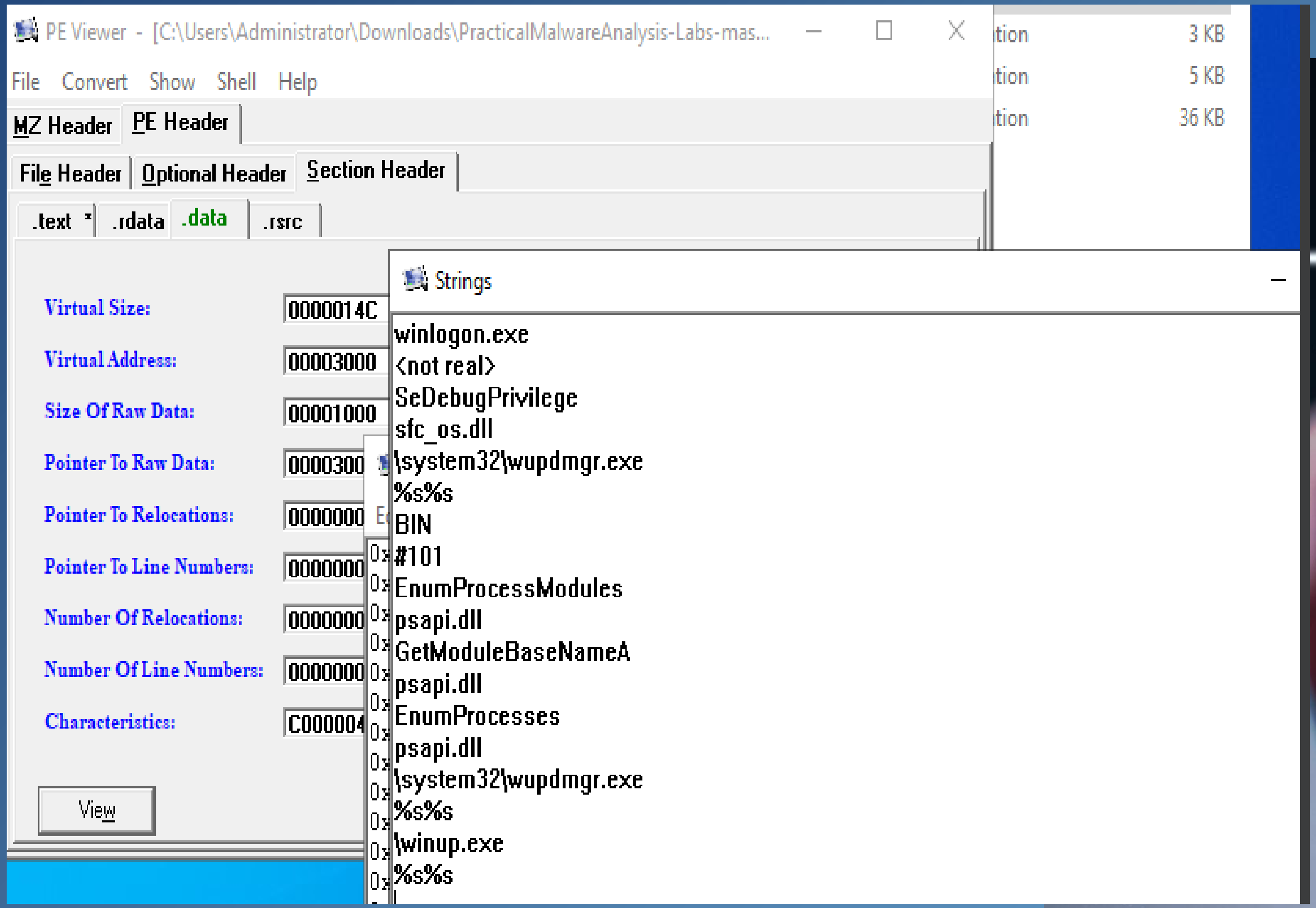


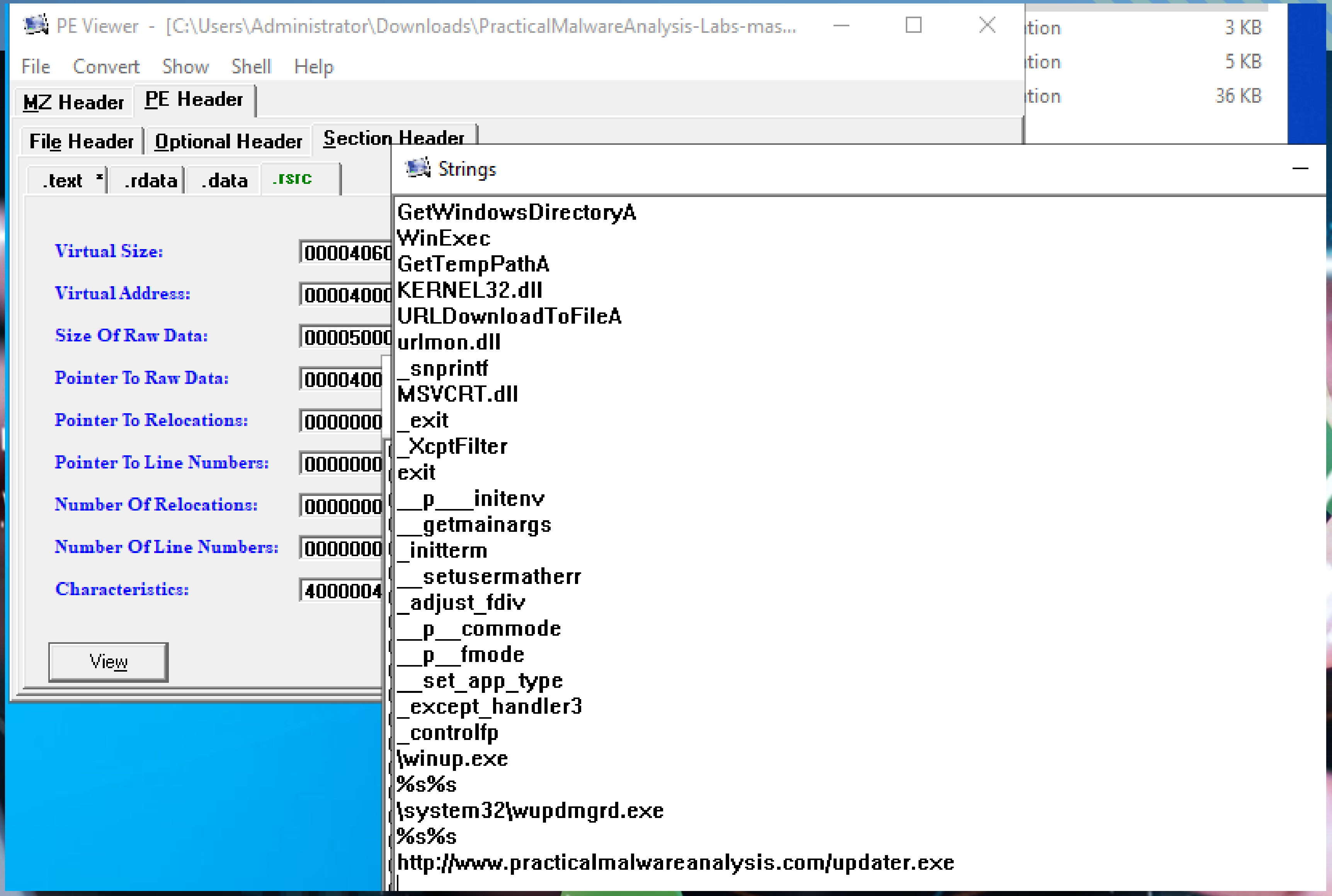
Step 2: Lab01-04





Cont. Step 2: Lab01-04



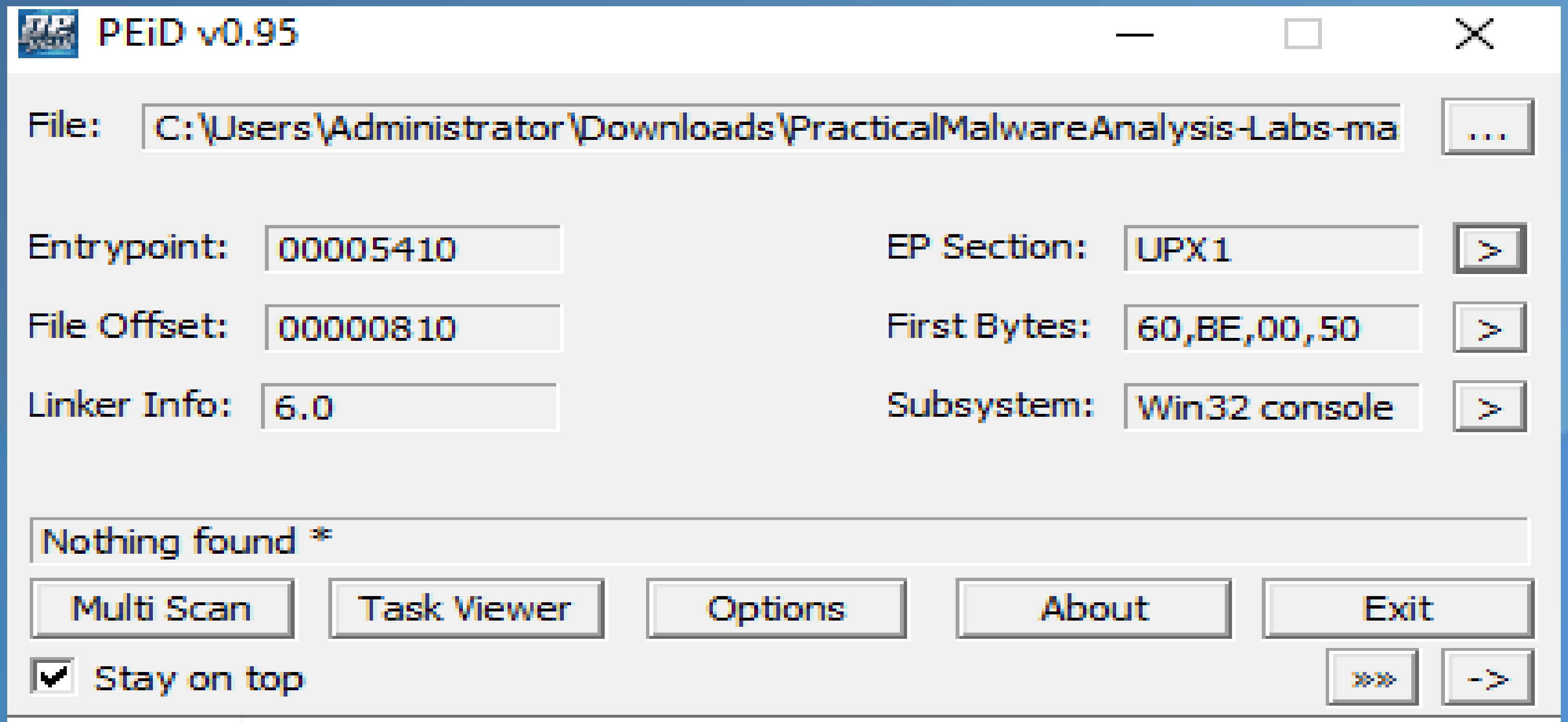


Step 3: Lab01-02

All the screenshots in Step 3 for the sample malware Lab02-04 are the results of using PEiD to find whether the malware was unpacked or

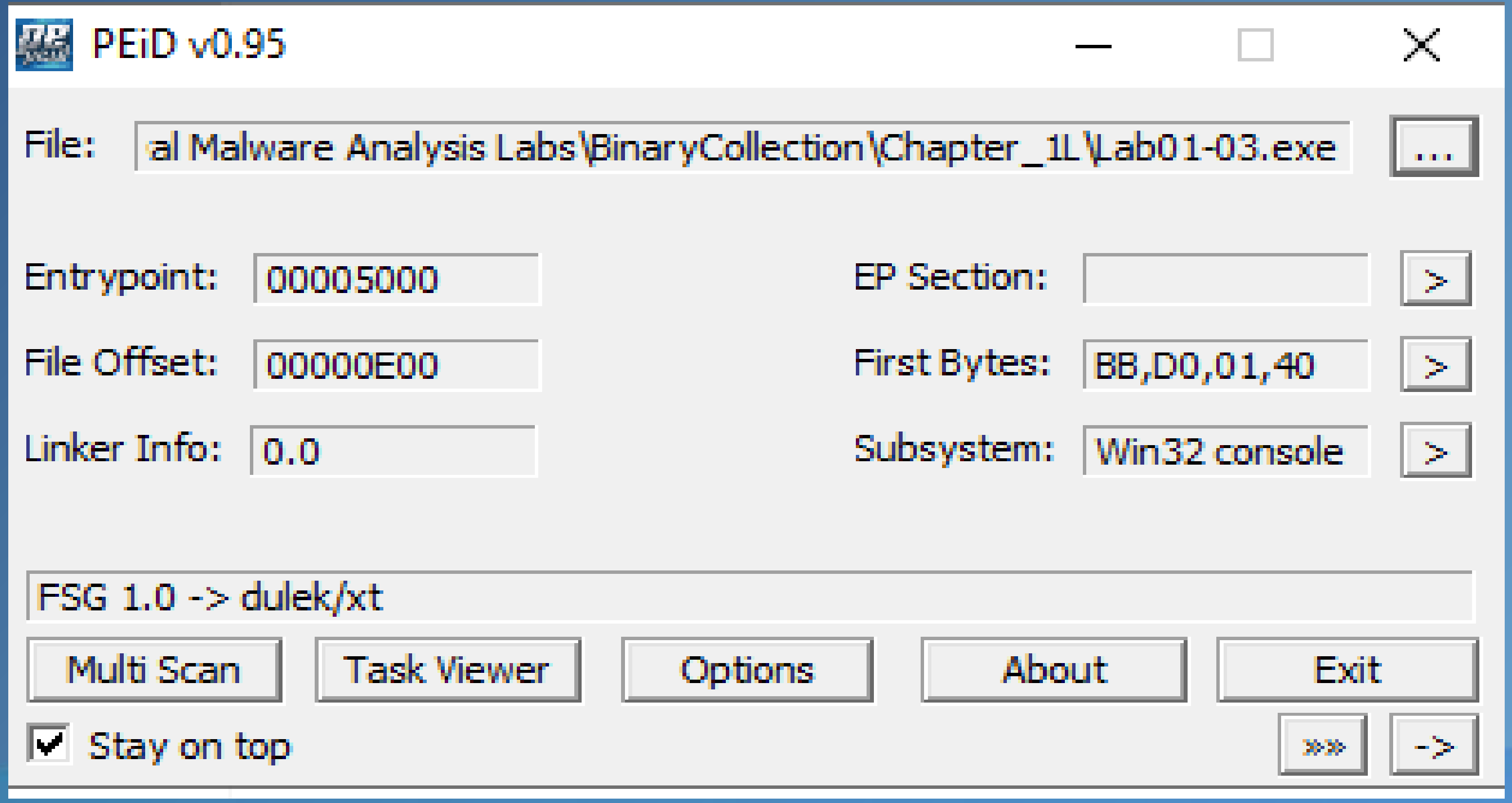
not.

The Lab01-02 malware is packed as the UPX1 in EP Section.



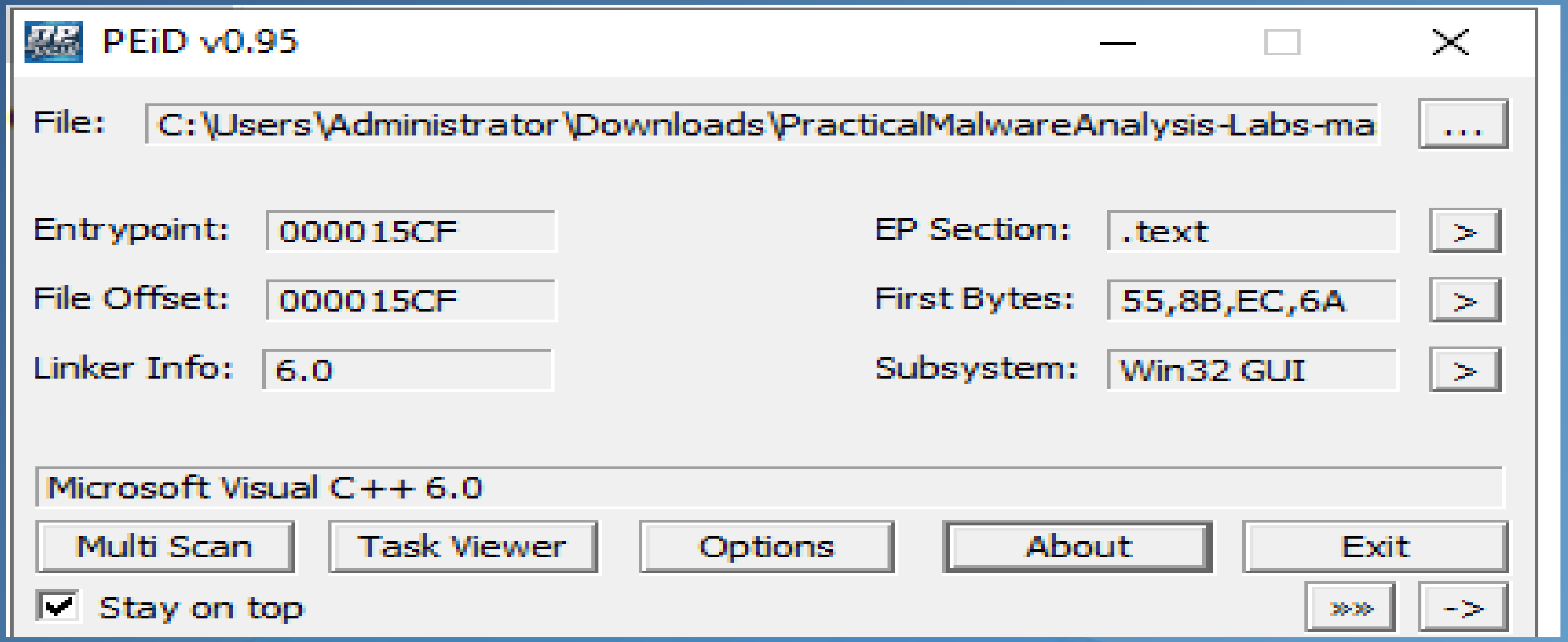
Step 3: Lab01-03

The Lab01-03 malware is corrupted / improperly packed or has a custom Packeron the file as it is blank in the EP Section.

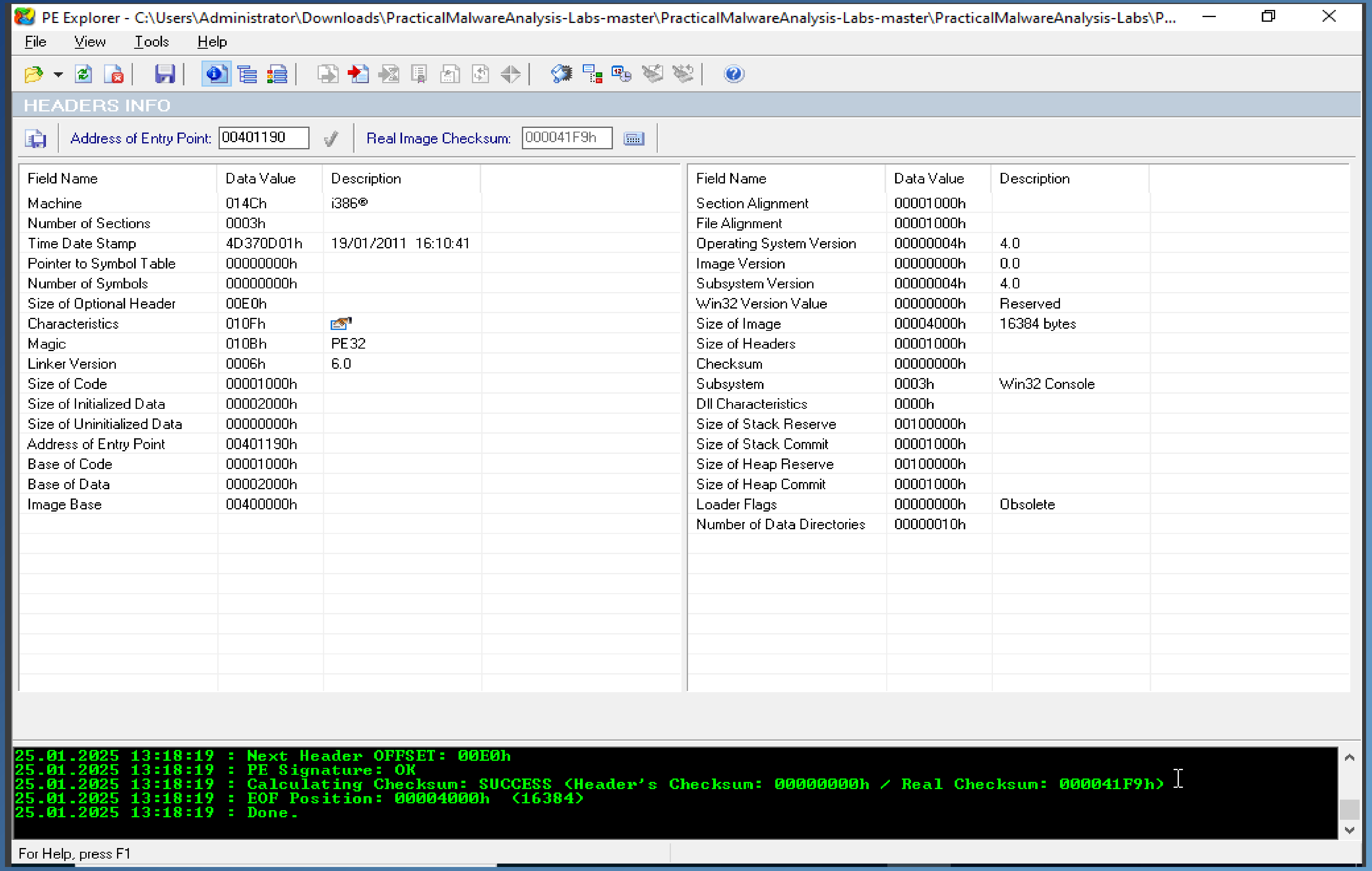


Step 3: Lab01-04

The Lab01-04 malware is unpacked as the .text in EP Section means that the file is still unpacked or no packing was discovered.

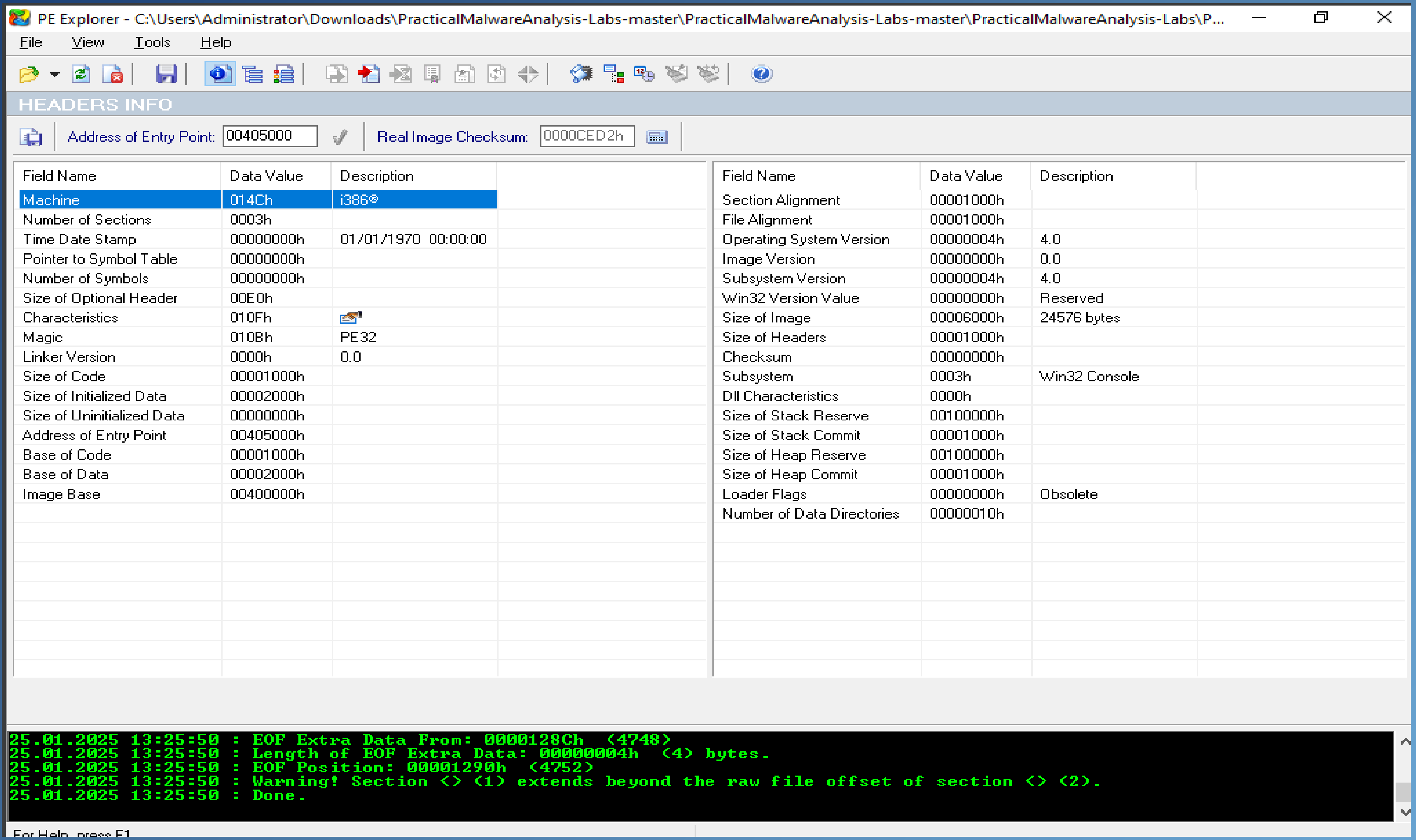


Step 4: Lab01-02



The Lab01-02 malware date and implementation were discovered using the Application PE Explorer.

Step 4: Lab01-03

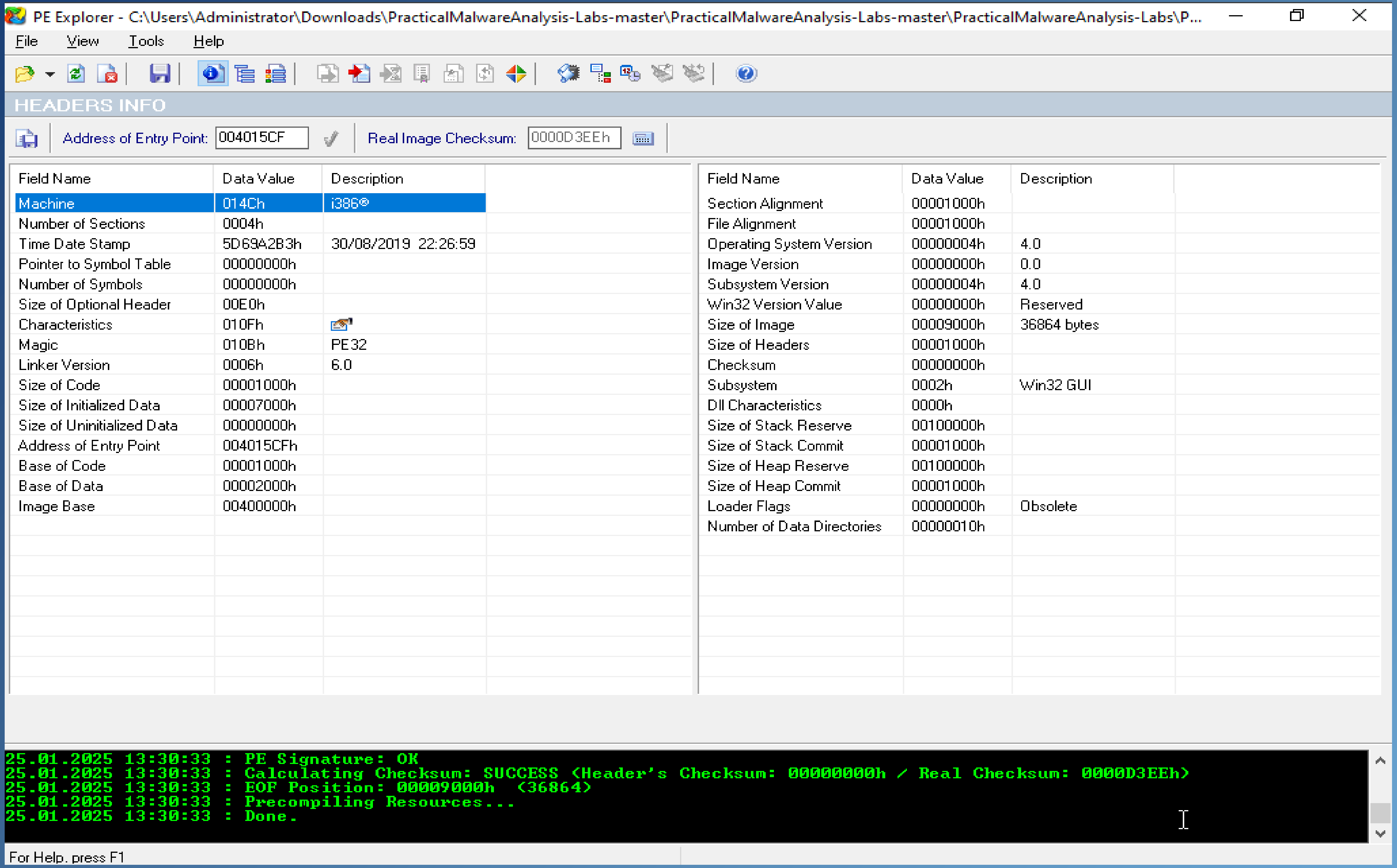


The Lab01-03 malware date and

implementation were discovered using the

Application PE Explorer.

Step 4: Lab01-04

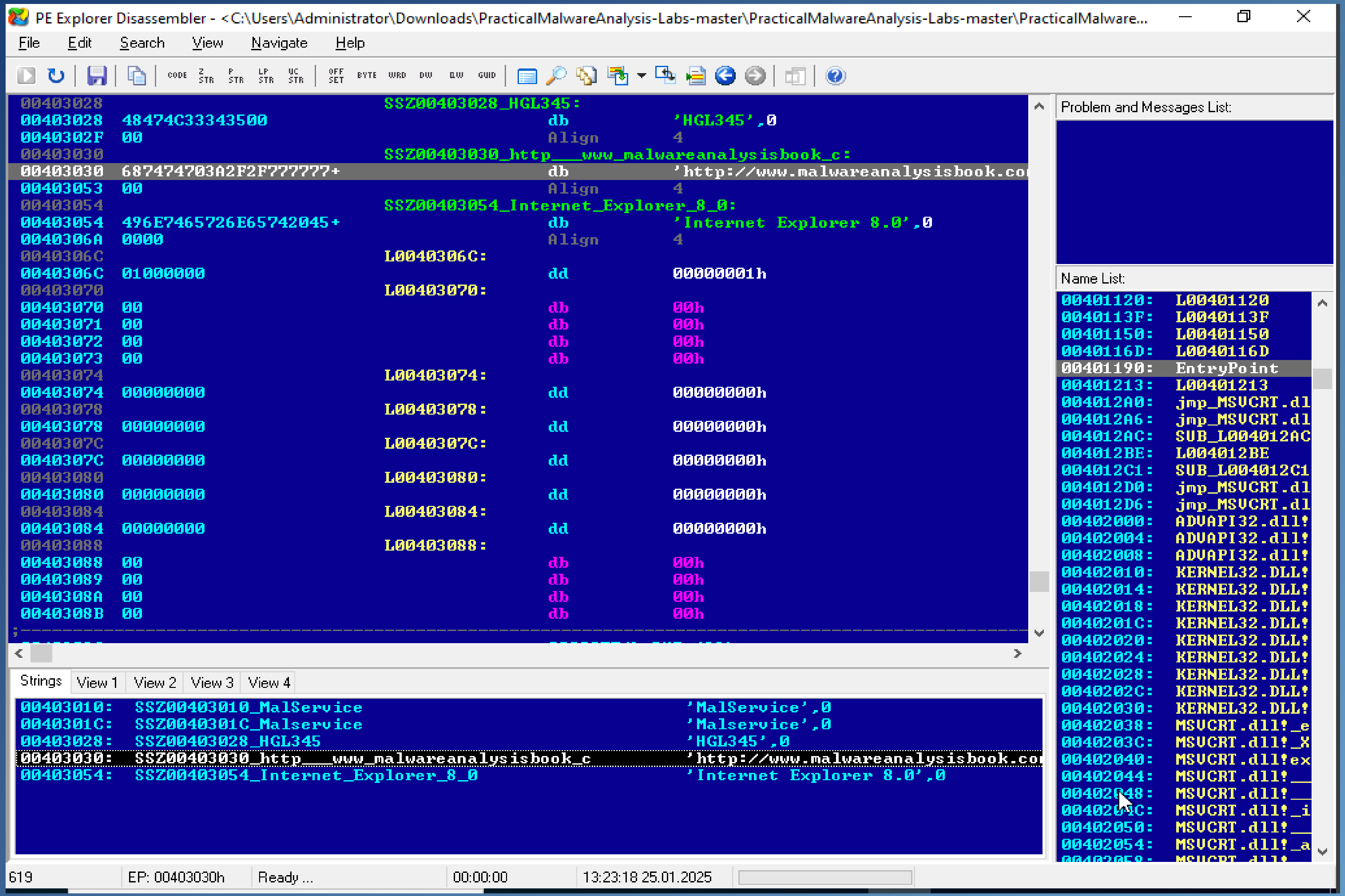


The Lab01-04 malware date and

implementation were discovered using the

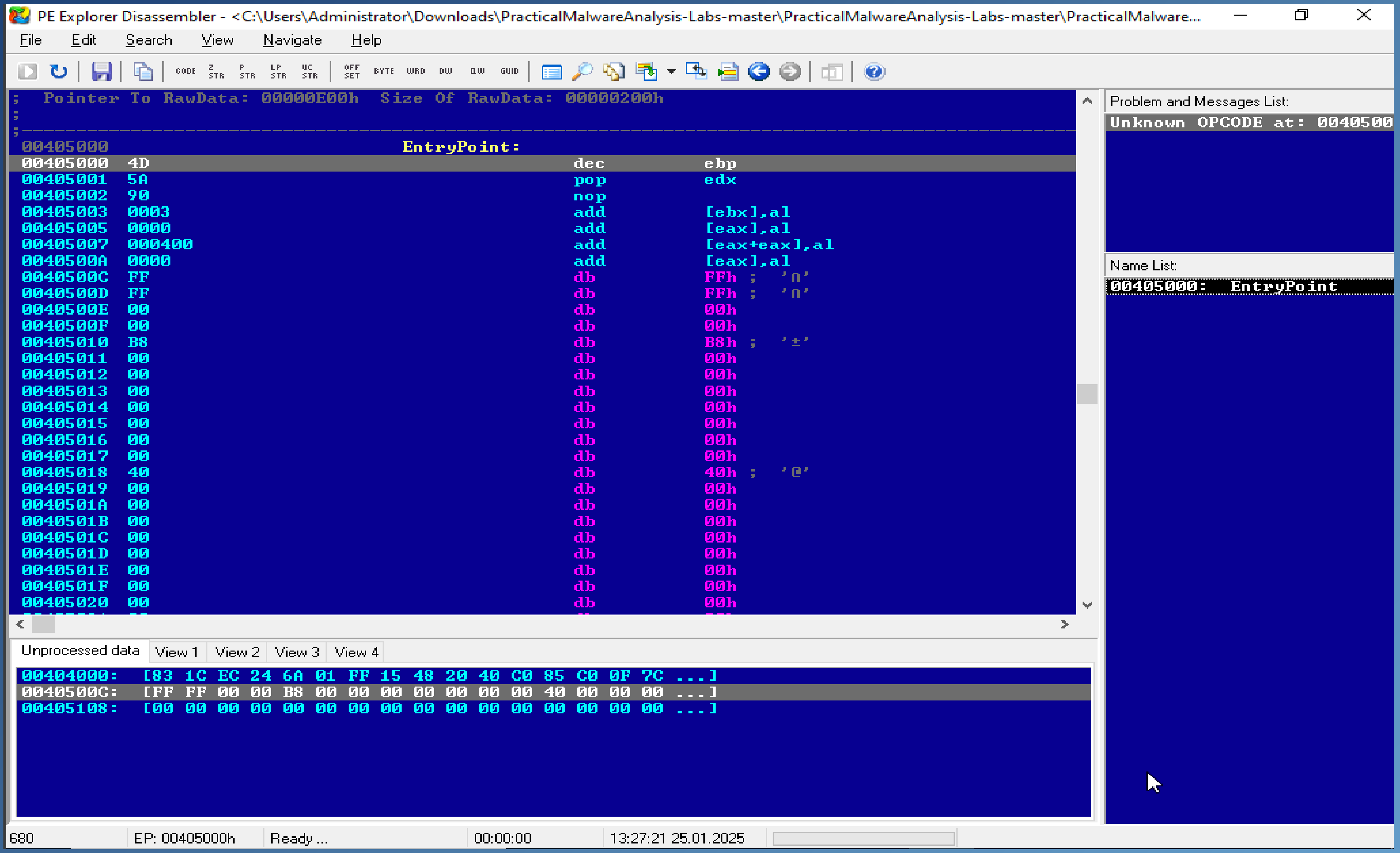
Application PE Explorer.

Step 5: Lab01-02



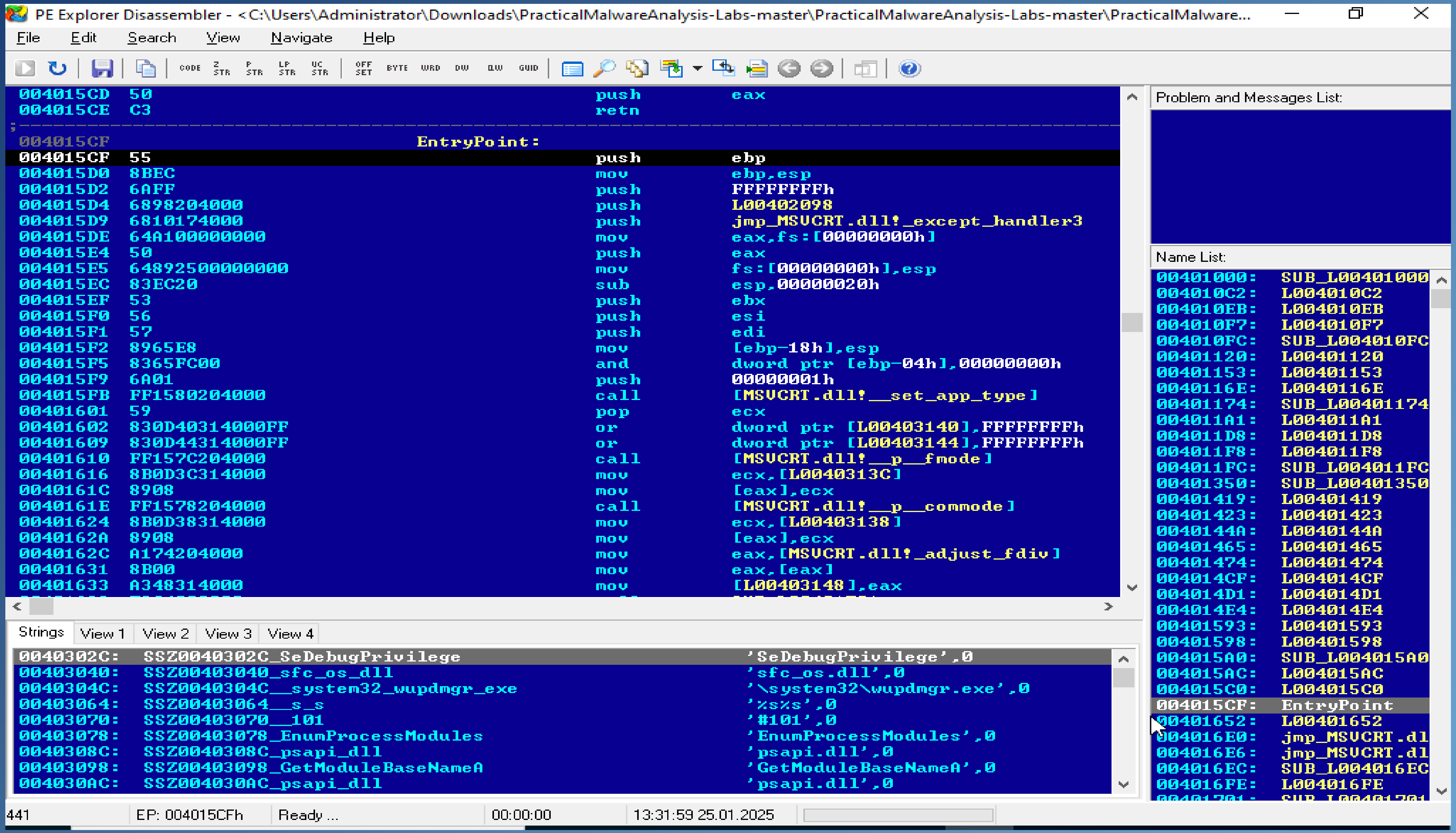
The Lab01-02 malware Disassembling results

Step 5: Lab01-03



The Lab01-03 malware Disassembling results

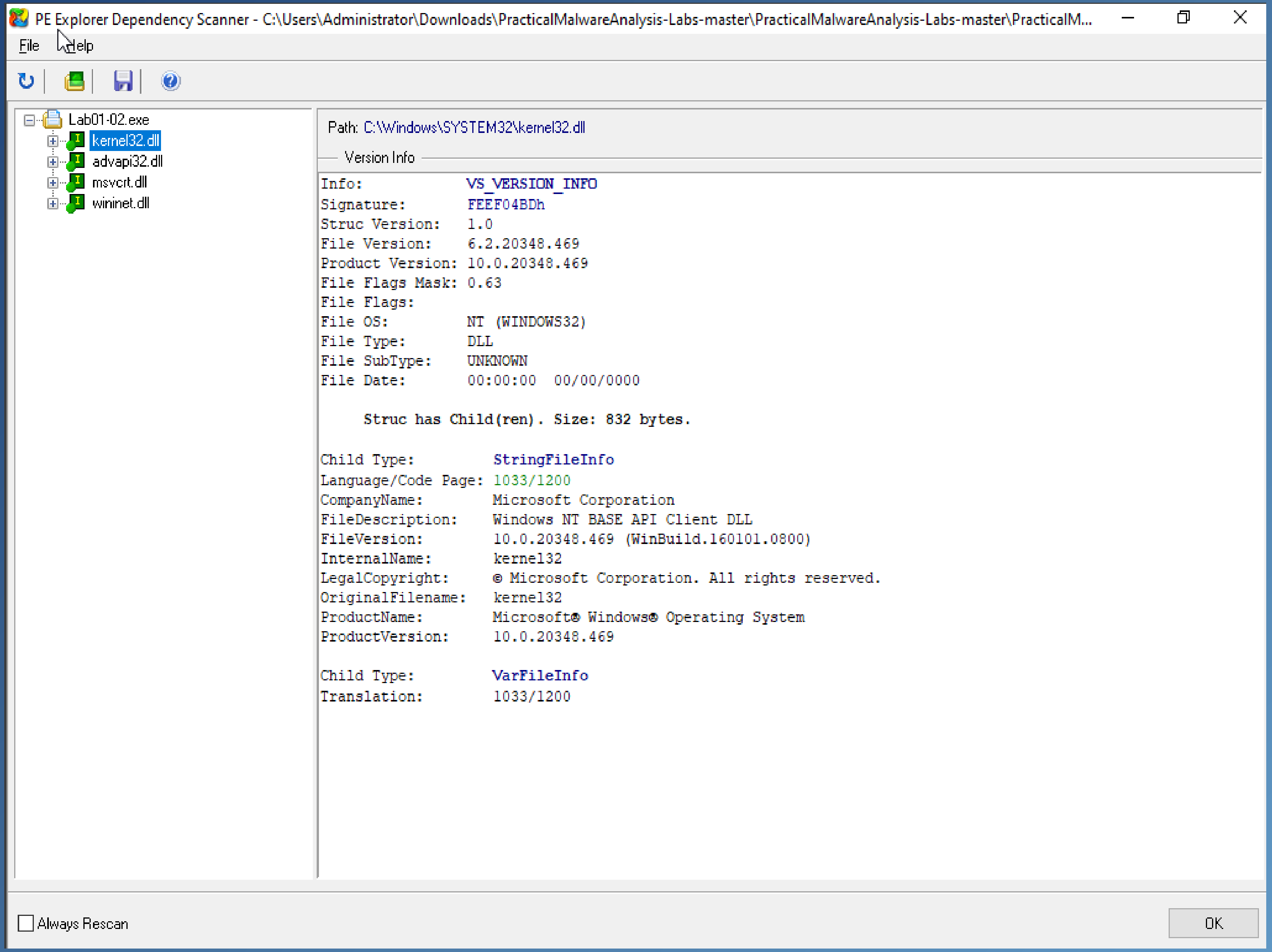
Step 5: Lab01-04

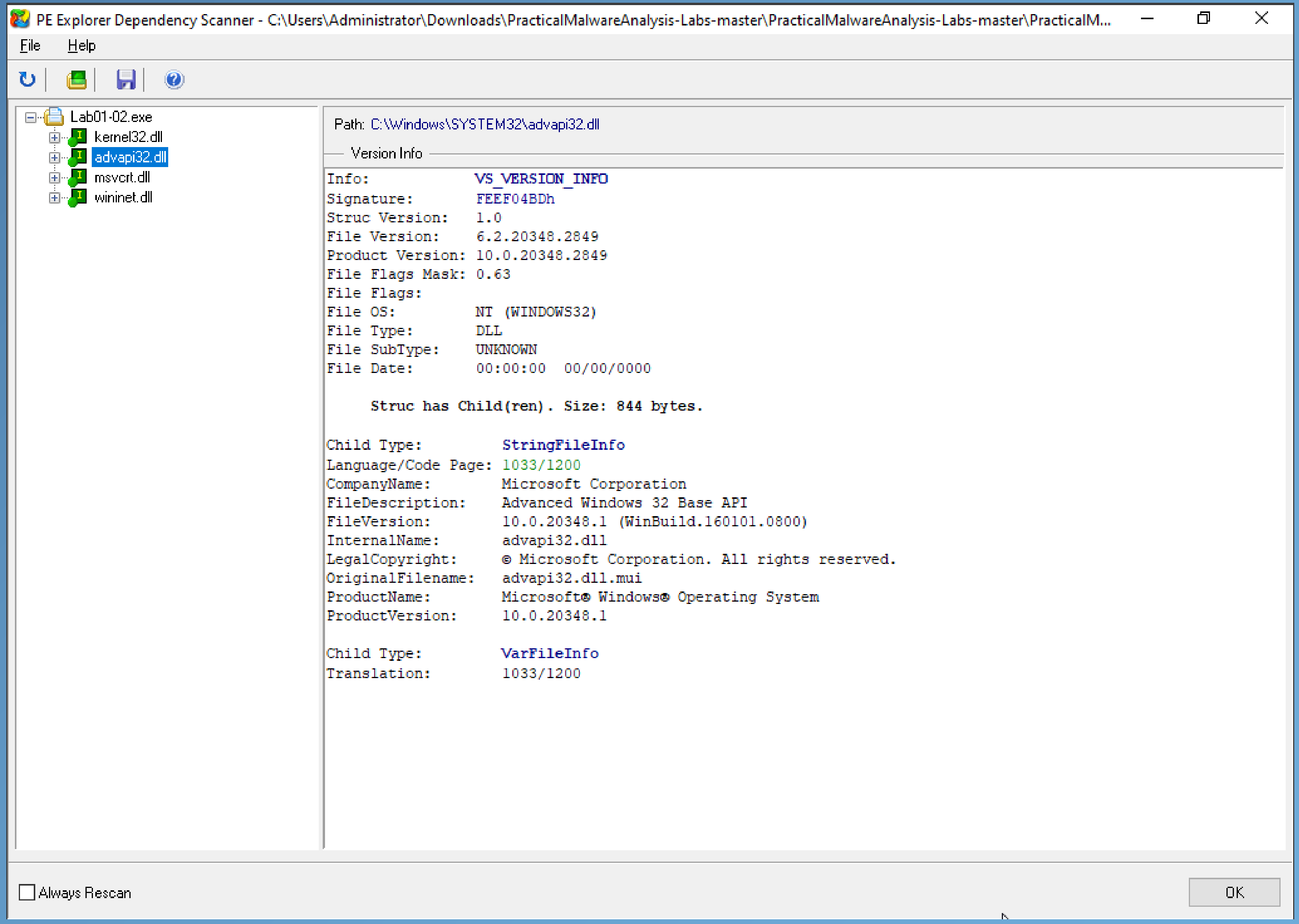


The Lab01-04 malware Disassembling results

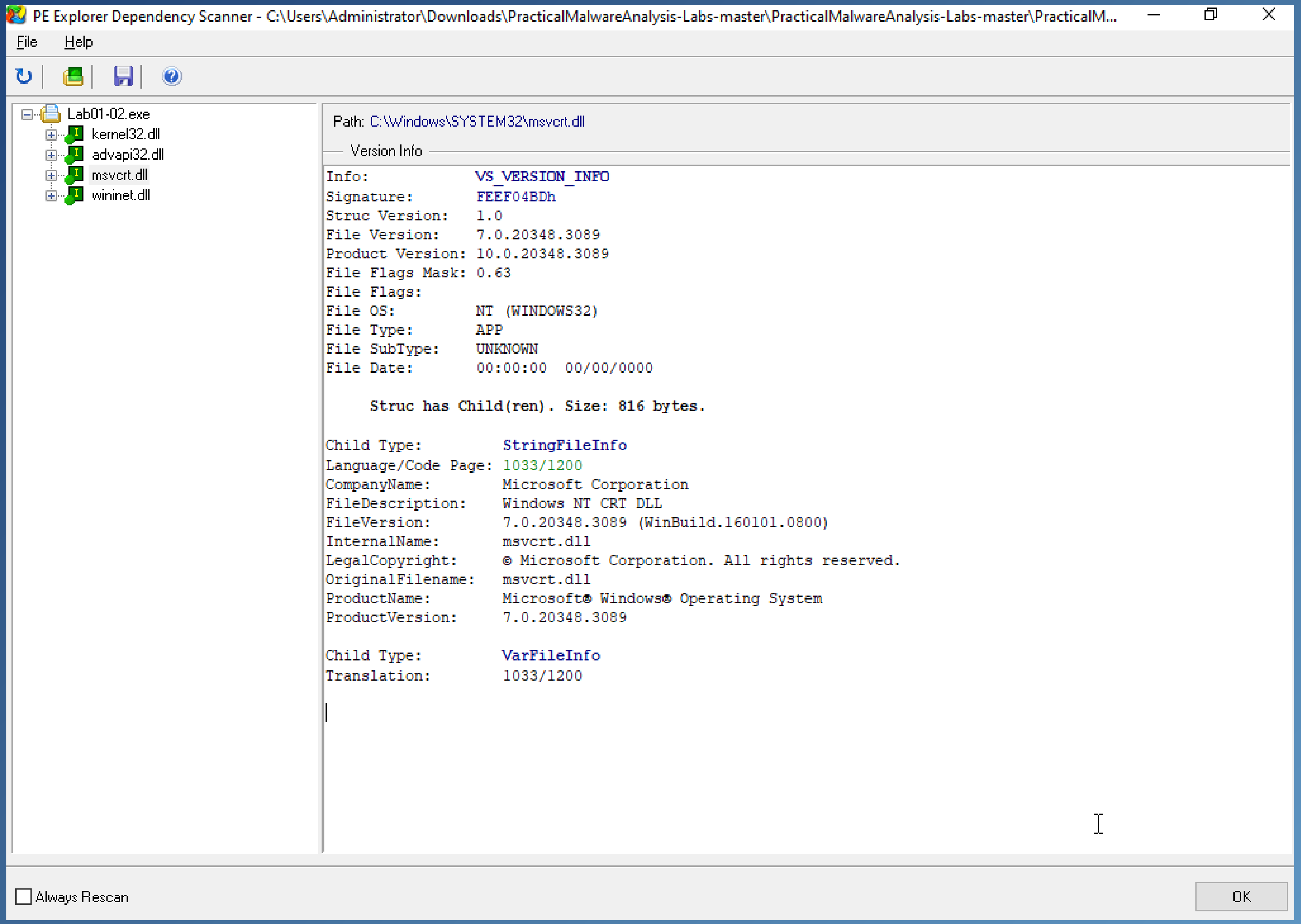
Step 6: Lab01-02

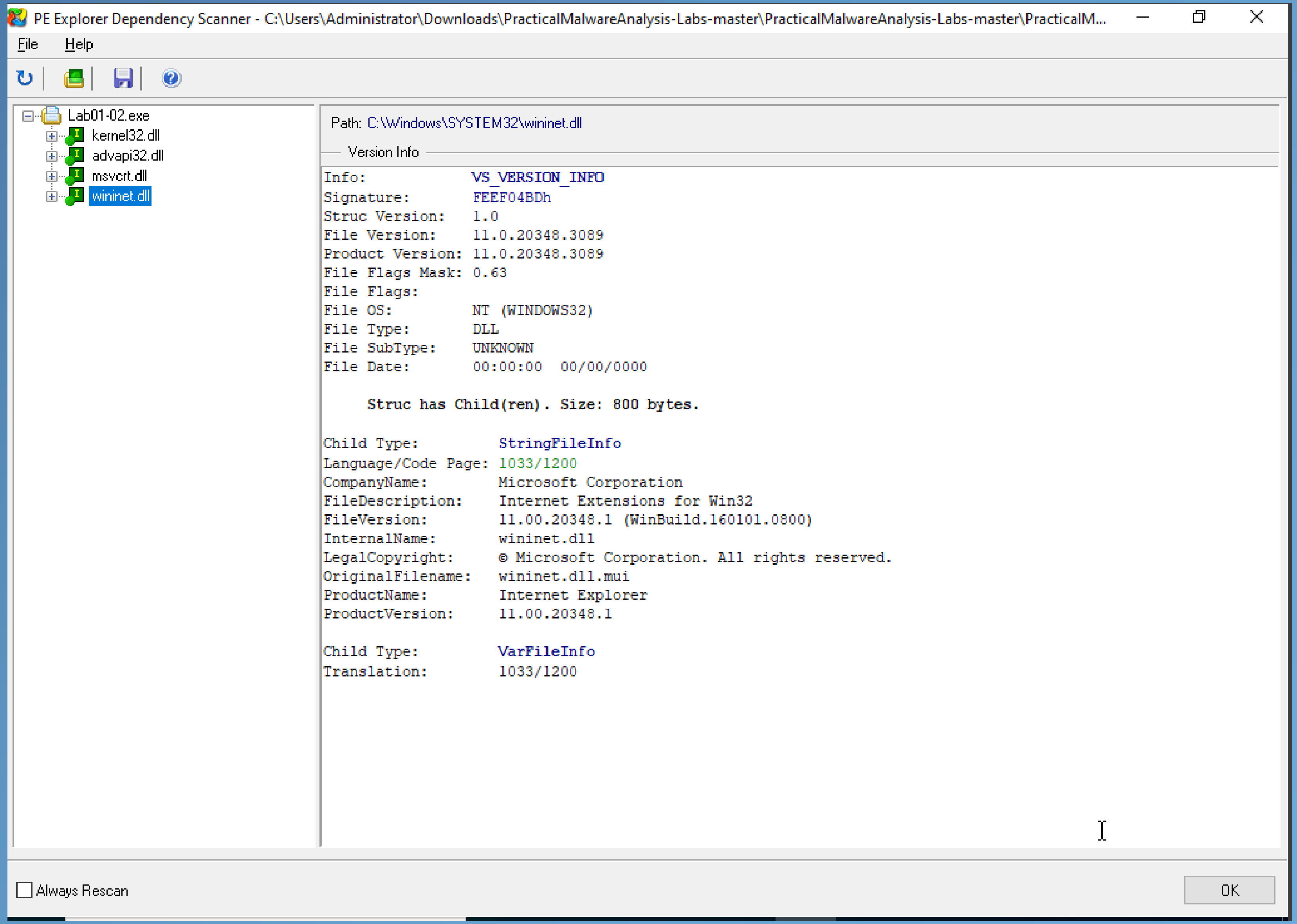
The Lab01-02 malware import functions are shown below



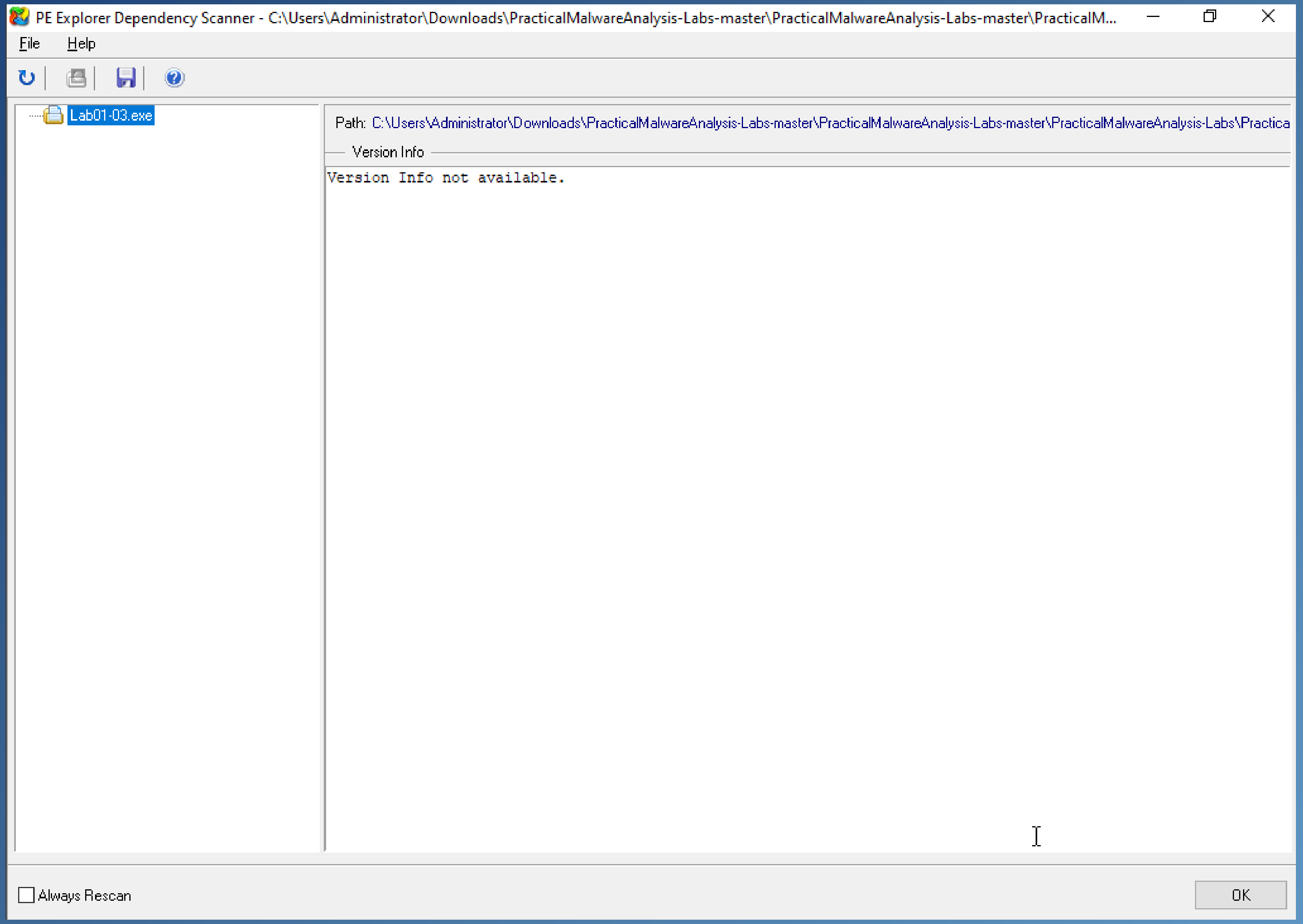


Cont. Step 6: Lab01-02





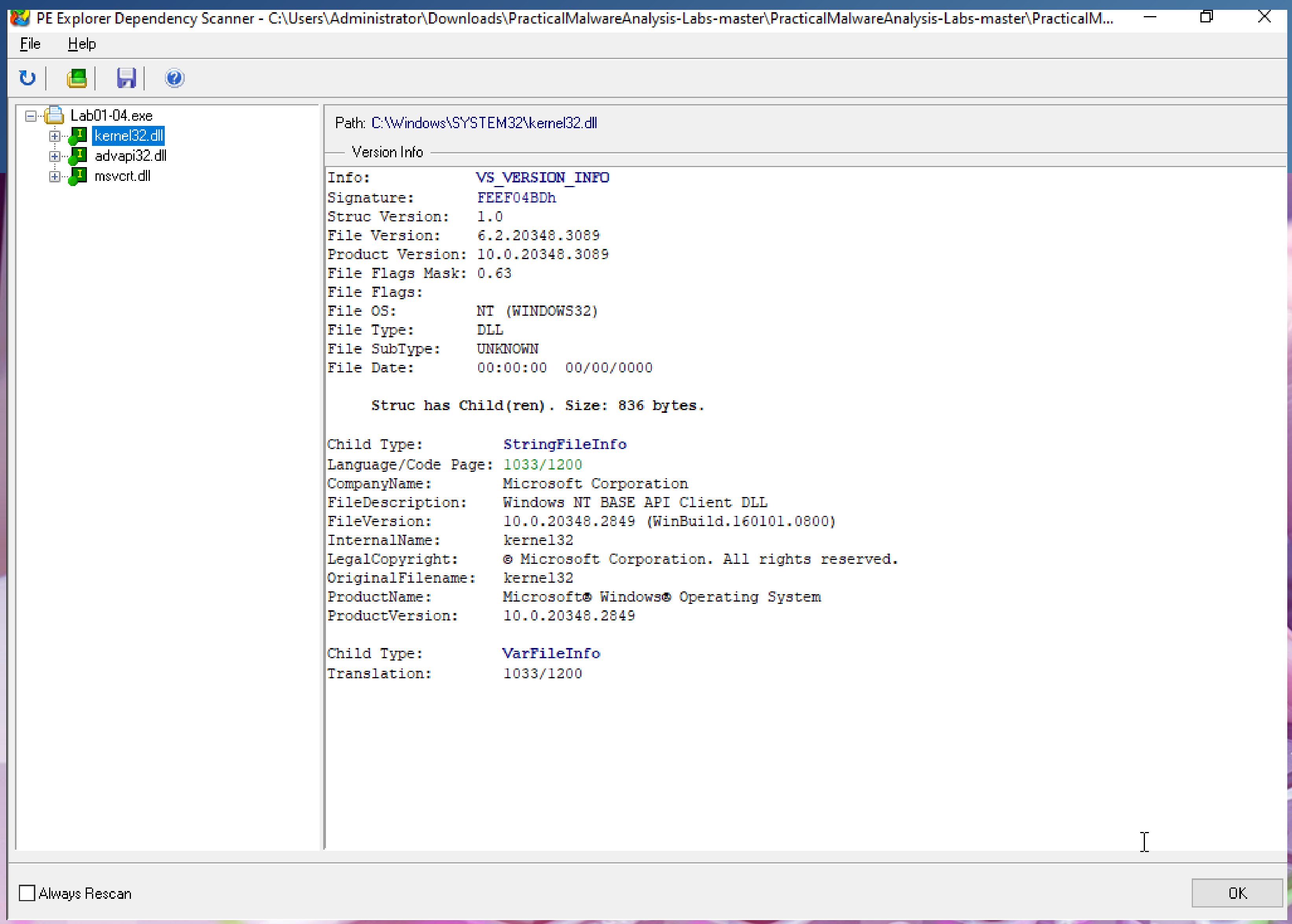
Step 6: Lab01-03

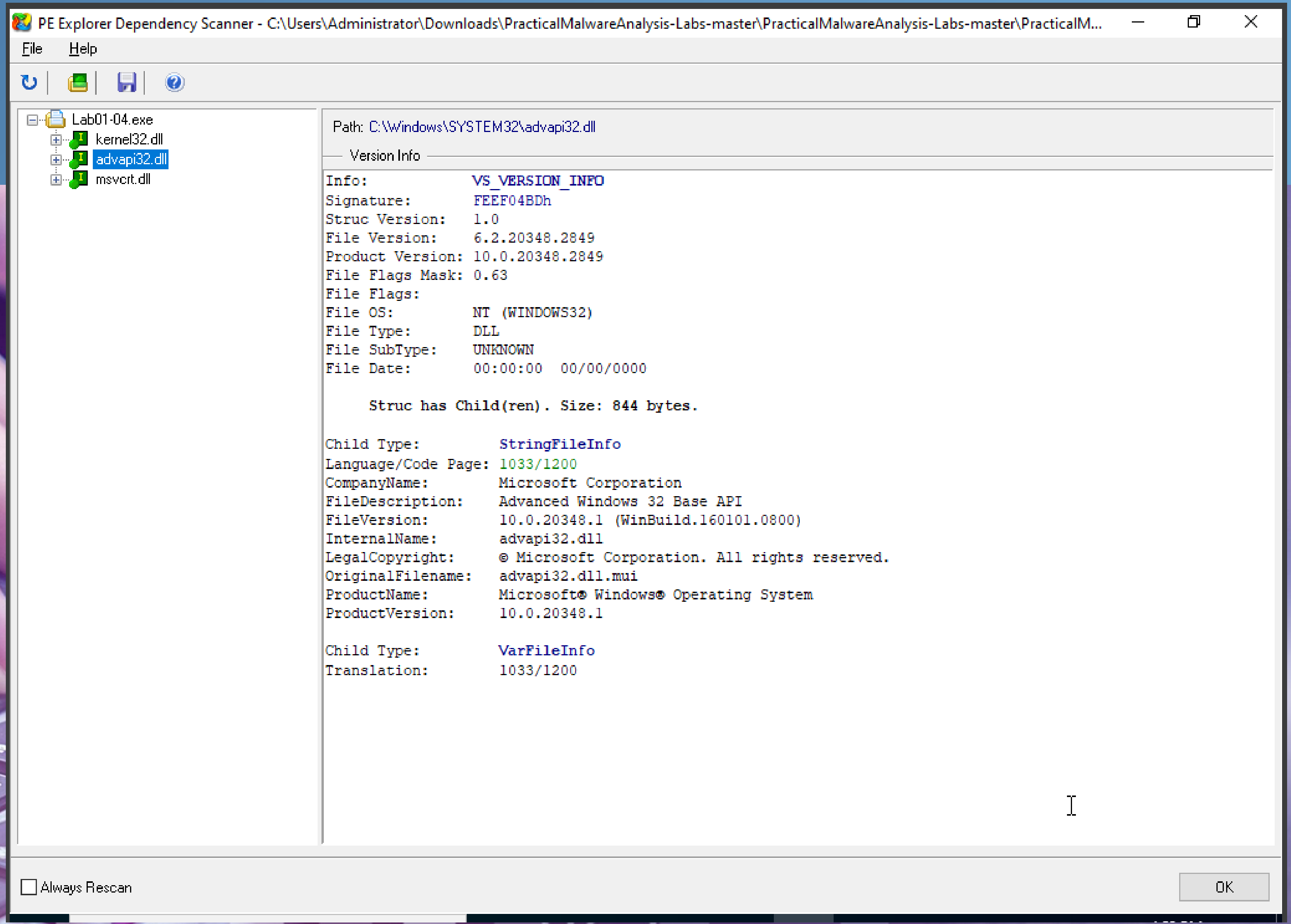


The Lab01-03 malware doesn't have any import functions attached to the file.

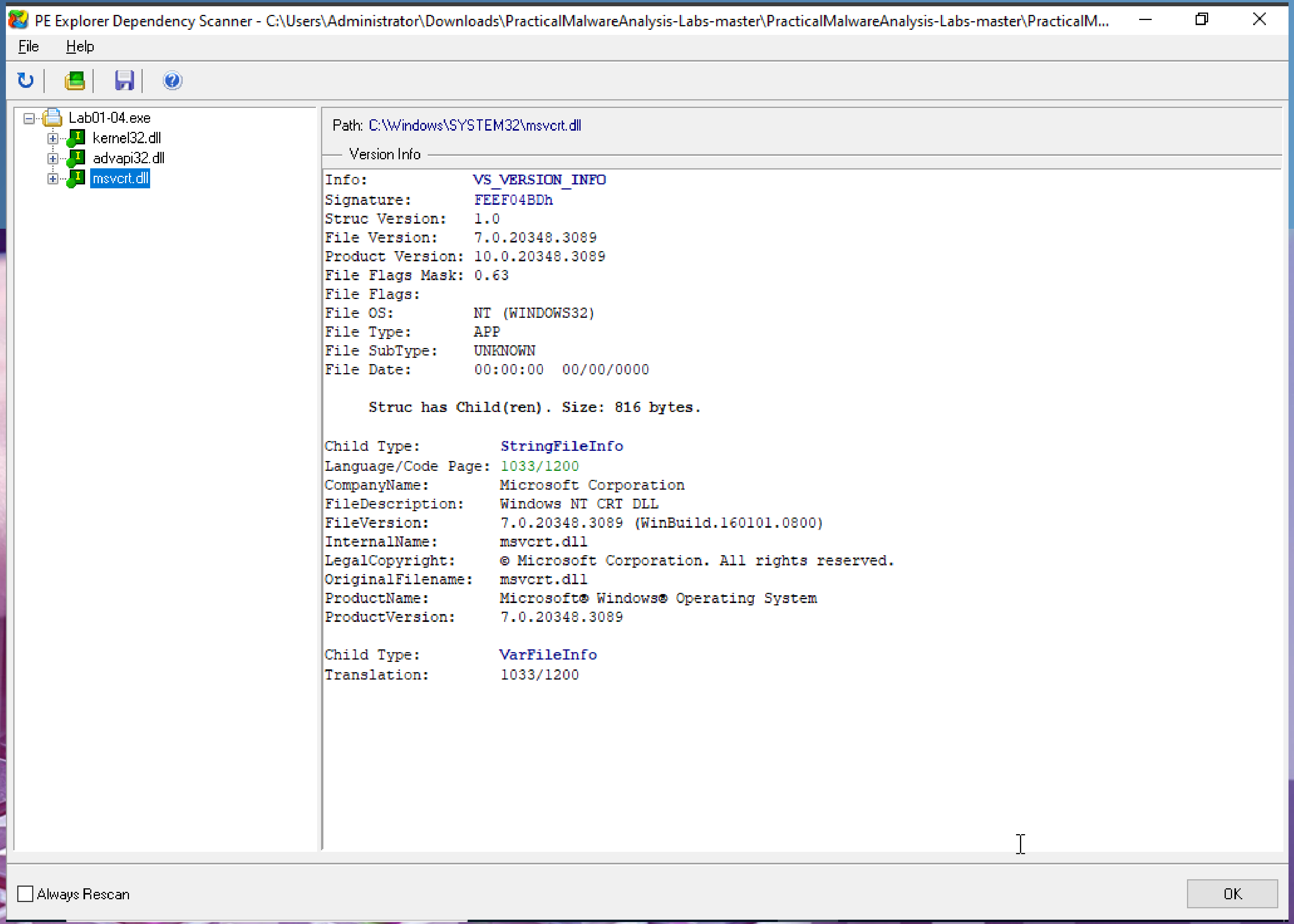
Step 6: Lab01-04

The Lab01-04 malware import functions are shown below





Cont. Step 6: Lab01-04



CONCLUSION

- In conclusion the project provides a structured and methodical approach to understanding malware through static analysis. By leveraging tools such as VirusTotal, PEViewer, PEiD, and PE-Explorer, the analysis process uncovered critical details about sample malware, including its packing status, metadata, string values, and import functions.
- The project highlighted the importance of thorough inspection and effective tool utilization in malware analysis, ensuring ethical and secure practices. It underscores the need for continuous learning in cybersecurity to counter evolving malware threats and protect digital systems

References

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- https://www.researchgate.net/figure/PEiD-tool-used-to-identify-if-the-malware-is-Packed-or-Not-Packed_fig6_367166446
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THANK YOU